Appendix C

Linear Algebra

Prerequisites for this chapter: Section A.1. Section A.2 is recommended.

C.1 Vector Spaces

Definition C.1.1 (Vector Space)

A *vector space* V over a field K, also called a K-vector space, is a set where you can add elements of V and also mutliply them by elements of K. More specifically, we have the following axioms.

1. associativity of addition: $(u+v)+w = u+(v+w)$ for any $u, v, w \in V$.
2. commutativity of addition: $u+v = v+u$ for any $u, v \in V$.
3. identity of addition: there is a $0_V \in V$ such that $u + 0_V = 0_V + u$ for any $u \in V$.
4. compatibility of multiplication: $(ab)v = a(bv)$ for any $a, b \in K$ and $v \in V$.
5. identity of multiplication: $1_K v = v$ for any $v \in V$ (1_K is the identity of K).
6. disitributy of multiplication: $a(u+v) = au + av$ and $u(a+b) = ua + ub$ for any $a, b \in K$ and $u, v \in V$.

The elements of the base field K are called *scalars* (and the elements of V *vectors*, although we won't use this termniology much).

These axioms are again all very obvious and you don't need to try to remember them, they are exactly the properties which let us establish the next propositions (that's why we defined it like that) so you just need to focus on what's next. As an example of vector spaces one can take the K-vector space K, the K-vector space K^n with componentwise addition and multiplication (these are regular vectors), $\mathbb{Z}/p^n\mathbb{Z}$ as a $\mathbb{F}_p$-vector space, and as a final more elaborate example the $\mathbb{R}$-vector space of functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(0) = 0$ (it's closed under addition).

When the base field is obvious from context or does not matter, we will drop the K.

Definition C.1.2 (Linear Independence)

We say elements $u_1, \ldots, u_n$ of a K-vector space V are *linearly independent* if no non-trivial linear combination of them is zero, i.e. $\sum_i a_i u_i = 0$ for $a_i \in K$ implies $a_i = 0$ for all i. Otherwise, we say they are *linearly dependent*.

For instance, the vectors $(1,0)$ and $(0,1)$ are linearly independent but the vectors $(1,0)$, $(0,1)$ and $(1,1)$ aren't because $(1,0) + (0,1) - (1,1) = 0$.

Definition C.1.3 (Bases)

A *K-basis* of a K-vector space V is a family of linearly independent elements $(e_i)_{i\in I}$ such that they span all of V: any element of V is a unique linear combination $\sum_i a_i e_i$ for some $a_i \in K$ (all but finitely many zero so that the sum makes sense).

The most common basis of $\mathbb{R}^n$ for instance is family set of unit vectors

$$(1, 0 \ldots, 0), (0, 1, \ldots, 0), \ldots, (0, \ldots, 0, 1).$$

The next proposition says that the cardinality of a basis (if there is one) does not depend on the basis, but only on the vector space. But first, here is an application (even though we have only made definitions!). When the base field K is obvious, we will drop the K and simply say "basis" and "vector space".

Problem C.1.1

There are $2n+1$ cows such that, whenever one exludes one of them, the rest can be divided into two groups of size n such that the sum of the weights of the cows in each group is the same. Prove that all cows have the same weight.

Solution

Let w_i be the weight of the ith cow. First, we solve the problem by induction when the weights are in $\mathbb{Z}$: if $\sum_{i\in I_k} w_i = \sum_{i\notin I_k, i\neq k} w_i$ then

$$\sum_{i=1}^{2n+1} w_i = w_k + \sum_{i\in I_k} w_i + \sum_{i\notin I_k, i\neq k} w_i \equiv w_k \pmod 2$$

so all weights have the same parity. If they are all even divide them by 2 and get a smaller solution (unless they are all equal to 0 in which case they are indeed equal), otherwise add 1 to all of them and divide them by 2; this yields another solution as the groups have the same size by assumption.

Now solve the problem over $\mathbb{Q}$: just multiply the weights by the lcm of the denominators to get weights in $\mathbb{Z}$ and we have already solved that case.

Finally, let's do the general case: $w_i \in \mathbb{R}$. Consider the $\mathbb{Q}$-vector space generated by the weights: $V = w_1\mathbb{Q} + \ldots + w_{2n+1}\mathbb{Q}$. Find a basis of V like this: pick any maximal subset of weights which are linearly independent $(w_i)_{i\in I}$. Indeed, any other weight w_k can be represented as a linear combination of $(w_i)_{i\in I}$ since there is a linear combination

$$aw_k + \sum_{i\in I} a_i w_i = 0$$

as $(w_k) \cup (w_i)_{i\in I}$ is linearly dependent by assumption, so

$$w_k = \sum_{i\in I} \frac{-a_i}{a} w_i$$

as $a \neq 0$ (otherwise $(w_i)_{i\in I}$ are linearly dependent).

Thus, we have found a basis $e_1, \ldots, e_m$ of V. To finish, write each w_i as $\sum_j a_{j,i} e_j$. We shall prove that $a_{j,1}, a_{j,2}, \ldots, a_{j,2n+1}$ satisfy the same conditions as the w_i (you can partition them into two groups of same sum and same size when you remove any one of them) for any j. Since

they are in $\mathbb{Q}$, by our prevous step this implies that they are all equal and thus the weights w_i are also all equal. This is however very easy: if $\sum_{i\in I} w_i = \sum_{i\in I'} w_i$ then

$$\sum_j e_j \sum_{i\in I} a_{j,i} = \sum_j e_j \sum_{i\in I'} a_{j,i}$$

so $\sum_{i\in I} a_{j,i} = \sum_{i\in I'} a_{j,i}$ by definition of a basis. We are done. ■

Now, we prove that any basis has the same cardinality if there exists a finite one. In that case we say the vector space is *finite-dimensional*. Unless otherwise stated, we will always work in the finite-dimensional case.

Proposition C.1.1 (Dimension)

Suppose $e_1, \ldots, e_n$ is a basis of a vector space V. Then, any basis of V has cardinality n. This n is called the *dimension* of V and written $\dim_K V$.

In fact we prove more.

Proposition C.1.2

Suppose $u_1, \ldots, u_n$ are linearly independent elements of V and $v_1, \ldots, v_m$ span all of V. Then $m \geq n$.

Since bases satisfy both of these conditions, we get $n \geq m$ and $m \geq n$ so $m = n$ for any bases of cardinality m and n.

Proof

We prove the contraposite: if $n > m$ then $u_1, \ldots, u_n$ are linearly dependent. Write $u_j = \sum_i a_{i,j} v_i$ with $a_{i,j} \in K$. We proceed by induction on m (when $m = 1$ it is obvious).

Pick an $a_{i,j} \neq 0$, this is possible otherwise all u_i are zero so in particular linearly dependent; without loss of generality assume $i = j = 1$. We will get rid of u_i and v_j for our induction. To do so, consider the family of vectors

$$u_1, u_2 - \frac{a_{1,2}}{a_{1,1}} u_1, u_3 - \frac{a_{1,3}}{a_{1,1}} u_1, \ldots, u_n - \frac{a_{1,n}}{a_{1,1}} u_1.$$

Clearly, these are linearly independent since you can recover $u_1, \ldots, u_n$ from them. Also,

$$u_j - \frac{a_{1,j}}{a_{1,1}} u_1 = \sum_{j\geq 2} (a_{i,j} - a_{1,i} a_{1,i} / a_{1,1}) v_i$$

has no coordinate in v_1. Thus by our induction hypothesis (e.g. on V' the space generated by $v_2, \ldots, v_m$) $u_2, \ldots, u_n$ are linearly dependent and we are done. (This idea of getting rid of coordinates will be used again in the proof of Theorem C.3.1, which states that the determinant of linearly independent vectors is non-zero.)

■

Here is a small application.

Problem C.1.2

Let $f, g \in \mathbb{R}[X]$ be two non-constant polynomials. Prove that there is an $h \in \mathbb{R}[X]$ such that fh is a polynomial in g.

Solution

This amounts to saying that some multiple of f is a polynomial in g. Let $n = \deg f$. We work in $\mathbb{R}[X]/f\mathbb{R}[X]$, i.e. $\mathbb{R}[X]$ modulo f. This an $\mathbb{R}$-vector space of dimension n as $(1, X, \dots, X^{n-1})$ is a basis.

Now consider the $n+1$ elements $1, g, g^2, \dots, g^n$, where g^k denotes the kth iterate of g. Since this family has more elements than the dimension of the vector space, they are linearly dependent: we get

$$\sum_i a_i g^i \equiv 0 \pmod{f}$$

for some not all zero $a_i \in \mathbb{R}$. This constitutes the wanted multiple of f. ■

Remark C.1.1

As this solution shows, the problem is extremely flexible. For instance, for any infinite set of non-negative integers S (e.g. the set of primes), f has a multiple whose non-zero monomials have the form X^s for some $s \in S$.

A final important result on basis, that we already saw in the proof of Problem C.1.1, is that we can always complete a family of linearly independent vectors to get a basis.

Proposition C.1.3

Any family of linearly independent vectors of a finite-dimensional vector space V can be completed into a basis by adding elements to it, and from any generating family we can extract a basis.

Exercise C.1.1*. Prove Proposition C.1.3.

As always, a quick application to finite fields. In Chapter 4 we prove that there exist fields of cardinality q for every prime power $q = p^n \neq 1$. Here we show the converse: if F is a field with q elements, then q is a prime power.

Proposition C.1.4

Suppose F is a field with q elements. Then there is some prime p and an integer n such that $q = p^n$.

Proof

The key is to consider F as a vector space over $\mathbb{F}_p$, where p is the characteristic of F. Indeed, the characteristic c is a prime, since if $0 = c = ab$ then either a or b must be zero too which means $a = c$ or $b = c$ by minimality of c.

Now F is a $\mathbb{F}_p$-vector space in the obvious way: just define $nx = \underbrace{x + \dots + x}_{n \text{ times}}$ (technically that's what we did before with the characteristic too) and this is compatible with $\mathbb{F}_p$ as $p = 0$ in F.

Since F is finite, it is also finite-dimension as a vector space: let $e_1, \dots, e_n$ be a basis (there exists one by Proposition C.1.3). Then, every element of F can be written in a unique way as

$$\sum_{i=1}^{n} a_i e_i$$

for some $a_i \in \mathbb{F}_p$. There are exactly p^n tuples $(a_1, \dots, a_n) \in \mathbb{F}_p$, so $q = p^n$ as wanted.

■

C.2 Linear Maps and Matrices

In this section, we consider morphisms of vector spaces which are called *linear maps*, or *linear transformations*.[1]

Definition C.2.1 (Linear Maps)

Let U and V be two K-vector spaces. A *linear map* $\varphi : U \to V$ is a function which is additive and homogeneous, i.e. $\varphi(x+y) = \varphi(x) + \varphi(y)$ and $\varphi(\lambda x) = \lambda\varphi(x)$ for any $x, y \in U$ and $\lambda \in K$.

For instance, the derivative map $f \mapsto f'$ is a linear map from $K[X]$ to itself (as K-vector spaces).

There is a very simple characterisation of linear maps $U \to V$. Let $u_1, \dots, u_m$ be a basis of U and $v_1, \dots, v_n$ of V. Write

$$\varphi(u_j) = \sum_i a_{i,j} v_i$$

for $j = 1, \dots, m$. Then f is uniquely defined from these $a_{i,j}$, and any system of $a_{i,j}$ gives rise to a linear map $U \to V$: if $x = \sum_j b_j u_j$ then

$$\varphi(x) = \sum_{i,j} b_j a_{i,j} v_i.$$

Note in particular that this shows that the structure of finite-dimensional vector spaces is more or less trivial: a vector space of dimension n is isomorphic to K^n. However, the profoundness of linear algebra lies precisely in what these isomorphisms are.

Remark C.2.1

When $K = \mathbb{Q}$, the $\mathbb{Q}$-linear maps are precisely the additive maps. Indeed, it follows from additivity that $\varphi(nx) = n\varphi(x)$ for $n \in \mathbb{Z}$, which implies that

$$\varphi\left(\frac{m}{n}x\right) = \frac{\varphi(mx)}{n} = \frac{m}{n}\varphi(x).$$

Additive functions are also called functions satisfying the "Cauchy equation" ($\varphi(x+y) = \varphi(x) + \varphi(y)$). This explains why this equation is unsolvable over $\mathbb{R}$: $\mathbb{R}$ is an infinite-dimensional $\mathbb{Q}$-vector space, so there are **a lot** of solutions: "just" pick a basis $(u_i)_{i \in I}$ of $\mathbb{R}$ and send u_i wherever you want. (It is however impossible, in the general case, to prove the existence of a basis of an infinite-dimensional vector space without the axiom of choice.)

Let us now comment a bit our proof of Lagrange's interpolation theorem A.1.2. What we did was consider the canonical basis $e_1, \dots, e_{n+1}$ where e_i has a 1 in the ith position and zeros everywhere else

[1] This shows that one should not call polynomial functions of degree 1 "linear", because they are not linear maps (unless the constant coefficient is 0)! One should call them "affine", because they correspond to affine transformations, not linear ones.

of the space K^{n+1} consisting of vectors $(b_1, \dots, b_{n+1})$. Then, for each element of this basis, we found polynomials f_i such that

$$(f_i(a_1), \dots, f_i(a_n)) = e_i.$$

Finally, we get the wanted result by taking linear combinations of these f_i since $e_1, \dots, e_{n+1}$ is a basis.

We come back to more abstract considerations. Given the bases $\mathcal{B} = (u_1, \dots, u_m)$ and $\mathcal{C} = (v_1, \dots, v_n)$, we denote the linear map φ in matrix form relative to the bases $\mathcal{B}$ and $\mathcal{C}$ by

$$M_{\mathcal{B}}^{\mathcal{C}}(\varphi) = \begin{bmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m,1} & a_{m,2} & \cdots & a_{m,n} \end{bmatrix}.$$

Note that we have used the index j for elements of the domain, and the index i for the codomain. This means that, to get the matrix of φ, we represent $\varphi(u_1), \dots, \varphi(u_m)$ by **column** vectors:

$$\begin{bmatrix} a_{1,1} \\ a_{2,1} \\ \vdots \\ a_{m,1} \end{bmatrix}, \dots, \begin{bmatrix} a_{1,n} \\ a_{2,n} \\ \vdots \\ a_{m,n} \end{bmatrix}$$

and then piece them together.

Definition C.2.2 (Matrices)

An $m \times n$ matrix is a family $(a_{i,j})_{i,j \in [m] \times [n]}$ which is denoted by $\begin{bmatrix} a_{1,1} & \cdots & a_{1,n} \\ \vdots & \ddots & \vdots \\ a_{m,1} & \cdots & a_{m,n} \end{bmatrix}$. The set of $m \times n$ matrices with coefficients in K is denoted by $K^{m \times n}$; when $n = 1$ we just write K^m.

Note that this last notation clashes with the Cartesian product, and that an element of K^m is a column vector not a row vector! To make things worse, we will even denote elements of K^m by $(a_1, \dots, a_m)$ as column vectors take too much place.

Here is how we define the product of two matrices A and B: if $A = M_{\mathcal{C}}^{\mathcal{B}}(\psi)$ and $S = M_{\mathcal{D}}^{\mathcal{C}}(\varphi)$ where $U \underset{\varphi}{\to} V \underset{\psi}{\to} W$, we want AB to correspond to $M_{\mathcal{D}}^{\mathcal{B}}(\psi \circ \varphi)$ where $D = (w_1, \dots, w_\ell)$ is a basis of W. Thus we compute

$$\psi(\varphi(u_j)) = \psi\left(\sum_k b_{k,j} v_k\right) = \sum_k b_{k,j} \psi(v_k) = \sum_k b_{k,j} \sum_i a_{i,k} w_i = \sum_i w_i \sum_k a_{i,k} b_{k,j}.$$

Hence we define $(a_{i,j})(b_{i,j}) = (c_{i,j})$ where $c_{i,j} = \sum_k a_{i,k} b_{k,j}$ (scalar product of the ith row of A with jth column of B). (In particular, the product of two matrices is only defined when the coordinates agree: $m \times n$ and $n \times \ell$.)

Definition C.2.3 (Matrix Multiplication)

The product of two matrices $(a_{i,j})$ and $(b_{i,j})$ of dimensions $m \times n$ and $n \times \ell$ is the matrix $(c_{i,j})$ of dimension $m \times \ell$ given by $c_{i,j} = \sum_k a_{i,k} b_{k,j}$.

Matrix multiplication is clearly associative, since composition is. It is however **not-commutative** in general. Similarly, addition of matrices is defined componentwise because we want $M_{\mathcal{C}}^{\mathcal{B}}(\varphi) + M_{\mathcal{C}}^{\mathcal{B}}(\psi)$ to correspond to $\varphi + \psi$. (We do not define multiplication of matrices to correspond to multiplication of linear maps because this does not make sense: $x \mapsto x \cdot x$ is not linear.)

Exercise C.2.1*. Prove that $\begin{bmatrix}1 & 0\\0 & 0\end{bmatrix}\begin{bmatrix}1 & 1\\0 & 0\end{bmatrix} = \begin{bmatrix}1 & 1\\0 & 0\end{bmatrix}$ but $\begin{bmatrix}1 & 1\\0 & 0\end{bmatrix}\begin{bmatrix}1 & 0\\0 & 0\end{bmatrix} = \begin{bmatrix}1 & 0\\0 & 0\end{bmatrix}$.

Exercise C.2.2*. Prove that matrix multiplication is *distributive* over matrix addition, i.e. $A(B+C) = AB + AC$ and $(A+B)C = AC + BC$ for any A, B, C of compatible dimensions.

Suppose we want to *invert* a linear map, i.e. find another linear map φ^{-1} such that $\varphi \circ \varphi^{-1} = \mathrm{id}$. The matrix of the identity is very simple to describe (with one basis): it's the matrix with ones on the diagonal and zero everywhere else

$$M_{\mathcal{B}}^{\mathcal{B}}(\mathrm{id}) = \begin{bmatrix}1 & 0 & \dots & 0\\0 & 1 & \dots & 0\\\vdots & \vdots & \ddots & \vdots\\0 & 0 & \dots & 1\end{bmatrix} := I_n$$

since

$$\mathrm{id}(e_j) = e_j = 0e_1 + \ldots + 0e_{j-1} + e_j + 0e_{j+1} + \ldots + 0e_n.$$

This matrix is called the *identity matrix*. Thus we would like to *invert* matrices. Why would this be useful? Well, this lets us, for instance, perform *changes of bases*: imagine that we first expressed φ with respect to the bases $\mathcal{B} = (u_1, \ldots, u_m)$ and $\mathcal{C} = (v_1, \ldots, v_n)$, but then we decided we actually preferred to express it with respect to $\mathcal{B}' = (u'_1, \ldots, u'_m)$ and $\mathcal{C}' = (v'_1, \ldots, v'_m)$. Consider the following two linear maps $\varphi_U(u_j) = u'_j$ and $\varphi_V(v_j) = v'_j$. It is clear that

$$M_{\mathcal{C}'}^{\mathcal{B}'}(\varphi) = M_{\mathcal{C}}^{\mathcal{B}}(\varphi_V \circ \varphi \circ \varphi_U^{-1})$$

since if $\varphi(u_j) = \sum_i a_{i,j} v_i$, then composing with φ_U^{-1} on the right transforms u_j into u'_j and composing with φ_V on the left transforms v_i into v'_i.

Thus,

$$M_{\mathcal{C}'}^{\mathcal{B}'}(\varphi) = M_{\mathcal{C}}^{\mathcal{B}}(\varphi_V) M_{\mathcal{C}}^{\mathcal{B}}(\varphi_U)^{-1}$$

(the matrices $M_{\mathcal{C}}^{\mathcal{B}}(\varphi_V)$ and $M_{\mathcal{C}}^{\mathcal{B}}(\varphi_U)^{-1}$ are called *change of bases matrices*). Of particular interest is the case where $U = V$, $\mathcal{B} = \mathcal{C}$ and $\mathcal{B}' = \mathcal{C}'$. In that case, we get an equality of the form $M' = NMN^{-1}$.

Finally, we prove one last result linking the surjectivity and injectivity of a linear map: if a linear map from a vector space to itself is injective, then it is surjective too and conversely. This is false in the infinite-dimension case, but this does not affect as as we only care about the finite-dimensional one. Recall what the kernel and image of a morphism are. Note that if φ is linear, then $\ker \varphi$ and $\mathrm{im}\, \varphi$ are vector spaces (as they are closed under addition, as well as multiplication by scalars).

Definition C.2.4 (Kernel and Image)

Let $\varphi : U \to V$ be a linear map. Its kernel $\ker \varphi$ is the set of $u \in U$ such that $\varphi(u) = 0$, and its image is the set of $v \in V$ such that $\varphi(u) = v$ for some $u \in U$.

Theorem C.2.1 (Rank-Nullity Theorem)

Suppose $\varphi : U \to V$ is a linear map (of finite-dimensional vector spaces). Then, $\dim \ker \varphi + \dim \mathrm{im}\, \varphi = \dim U$.

Remark C.2.2

This is called the "rank-nullity theorem" because dim im is also called the rank, and nullity means dim ker.

Proof

Let $u_1, \dots, u_k$ be a basis of $\ker \varphi$ and $\varphi(u'_1), \dots, \varphi(u'_m)$ a basis of $\operatorname{im} \varphi$. We prove that $u_1, \dots, u_k, u'_1, \dots, u'_m$ is a basis of U.

First we prove that these elements are linearly independent. Suppose that

$$\sum_i a_i u_i + \sum_i b_i u'_i = 0.$$

Then, $\sum_i b_i f(u'_i) = 0$ by composing with f. Since $f(u'_1), \dots, f(u'_m)$ are linearly independent, this means $b_1 = \dots = b_m = 0$. Then, from $\sum_i a_i u_i = 0$, we deduce $a_i = 0$ since $u_1, \dots, u_k$ are also linearly independent.

It remains to prove that they span all of U. Let $u \in U$ be an element. Write $f(u) = \sum_i b_i f(u'_i)$. Then, $f\left(u - \sum_i b_i u'_i\right) = 0$. This means $u - \sum_i b_i u'_i \in \ker f$, so

$$u - \sum_i b_i u'_i = \sum_i a_i u_i \iff u = \sum_i a_i u_i + \sum_i b_i u'_i$$

as wanted.

■

Corollary C.2.1*

A linear map $U \to U$ is injective if and only if it is surjective. In other words, a square matrix has a right-inverse if and only if it has a left-inverse.

Proof

A linear map is injective if and only if its kernel is trivial, i.e. has dimension 0. By the rank-nullity theorem, this is equivalent to $\dim \operatorname{im} \varphi = \dim U$, i.e. φ being surjective.

A $n \times n$ matrix A has a right-inverse if and only if the linear map from $K^{n \times n} \to K^{n \times n}$ defined by $B \mapsto AB$ is surjective, and if A^{-1} is this inverse then $A(A^{-1}A) = A$ so $A^{-1}A = I_n$ by injectivity (from the rank-nullity theorem) which means A^{-1} is a left inverse too. But if A has a left-inverse, then $B \mapsto AB$ is injective which means it's surjective too.

■

Another corollary is that this lets us deduce the existence part of the Lagrange interpolation theorem (Theorem A.1.2) from the uniqueness part: for any $a_1, \dots, a_n \in K$, the map from the vector space $K_n[X]$ of polynomials of degree less than n to K^n given by

$$f \mapsto (f(a_1), \dots, f(a_n))$$

is injective so must be surjective too since $K_n[X]$ and K^n both have dimension n.

Here is a combinatorial application of the fact that the right-inverse of a matrix is also its left-inverse.

Problem C.2.1

There are $2n$ boys and $2n$ girls at a party. For each pair of girls, there are exactly n boys that danced with exactly one of them. Prove that the same is true if we exchange the words "boys" and "girls" in the last sentence.

Solution

Consider the *adjacency matrix* $M = (a_{i,j})$ defined by $a_{i,j} = 1$ if the ith girl and the jth boy have danced together and -1 otherwise (so the rows correspond to the girls and the columns to the boys). We claim that the condition of the problem is exactly equivalent to

$$MM^\mathsf{T} = 2nI_{2n},$$

where M^T designates the *transpose matrix* of M, i.e. the matrix $M^\mathsf{T} = (b_{i,j})$ where $b_{i,j} = a_{j,i}$ (exchange the rows with the columns). Let's compute this product: the (i, j) coordinate is

$$c_{i,j} = \sum_k a_{i,k}b_{k,j} = \sum_k a_{i,k}a_{j,k}.$$

Now what is $a_{i,k}a_{j,k}$? $a_{i,k}$ corresponds to whether the ith girl has danced with the kth boy, and $a_{j,k}$ to whether the jth girl has danced with the kth boy. Thus $a_{i,k}a_{j,k}$ is $-1 = 1 \cdot (-1) = (-1) \cdot 1$ if exactly one of them has danced with him, and $1 = 1 \cdot 1 = (-1)(-1)$ otherwise.

We conclude that the sum $\sum_k a_{i,k}a_{j,k}$ is zero if and only if there are exactly n boys which danced with exactly one of the girls i, j: for $i \neq j$ this is exactly what the problem says! For $i = j$ the sum is trivial: $a_{i,k}^2 = 1$ so $c_{i,j} = \sum_k 1 = 2n$. We have thus proven our claim: the condition is equivalent to $MM^\mathsf{T} = 2nI_{2n}$, i.e. that $M^\mathsf{T}/2n$ is the right-inverse of M. But then $M^\mathsf{T}/2n$ is also the left-inverse of M by the rank-nullity theorem, so

$$\frac{1}{2n}M^\mathsf{T}M = I_{2n} \iff M^\mathsf{T}(M^\mathsf{T})^\mathsf{T} = 2nI_{2n}$$

which means that the statement is true with the word "boys" and "girls" exchanged (since this is what the transpose does: it exchanges the rows with the columns). ■

C.3 Determinants

The set of matrices almost form a **non-commutative** ring under addition and multiplication, except that multiplication is not always defined (only when the dimensions are compatible). However, for square matrices it is always possible. Thus, square matrices are usually nicer to study, for instance we have seen that they have a right-inverse and if and only they have a left-inverse, which is not true for other matrices (by the rank-nullity theorem). In this section, we find a criterion to determine those square matrices are invertible. Note that finding when a $n \times n$ matrix M is invertible is equivalent to finding when the rows or the columns are linearly independent. Indeed, the column are linearly independent if and only if the images of the canonical basis

$$e_1 = (1, 0, \dots, 0), \dots, e_n = (0, \dots, 0, 1)$$

by the linear map $v \mapsto Av$ from K^n to K^n are linearly independent, i.e. if and only if this map is injective (which is equivalent to A being invertible). For the rows, one can consider the map $v \mapsto A^\mathsf{T}v$, as $(AB)^\mathsf{T} = B^\mathsf{T}A^\mathsf{T}$ so A is invertible if and only if its transpose is.

Exercise C.3.1. Prove that an $m \times n$ matrix can only have a right-inverse if $m < n$, and only a left-inverse if $m > n$. When does such an inverse exist?

Exercise C.3.2*. Prove that $(AB)^\mathsf{T} = B^\mathsf{T}A^\mathsf{T}$ for any $n \times n$ matrices A, B.

Let's start with the 2×2 case. Let $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ be such a matrix. The vectors (a, b) and (c, d) are linearly dependent if and only if there are some x, y such that $(ax + cy, bx + dy) = (0, 0)$. By rescaling x and y if necessary, we may assume $x = -c$ and $y = a$ from $ax + cy = 0$ (unless $a = c = 0$ but in that case they're clearly linearly dependent). Then, $bx + dy = 0$ becomes $ad - bc = 0$.

Thus, $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is invertible if and only if $ad - bc \neq 0$. In fact, we can even check that

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = (ad - bc) I_2.$$

Exercise C.3.3. Prove this identity.

This number $ad - bc$ is called the *determinant* $\det M$ of the matrice $M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Our goal will be to define such a determinant for $n \times n$ matrices satisfying the following properties:

- $\det M$ is a homogeneous polynomial of degree n in the coordinates of M. Moreover, $\det M$ has degree 1 in each coordinate of M.
- M is invertible if and only if $\det M \neq 0$.

In fact, these properties define uniqueley the determinant up to leading coefficient (this will be proven in Theorem C.3.4)!For the sake of convenience, since we are going to be talking more about columns and rows, we denote the ith row of M by M_i and the jth column of M by M^j so that

$$M = [M^1, \cdots, M^n] = \begin{bmatrix} M_1 \\ \vdots \\ M_n \end{bmatrix}.$$

Here is how one can define the determinant of $n \times n$ matrices inductively. In Theorem C.3.4 and Theorem C.3.3, we will give another characterisations of the determinant, the last one being completely explicit. The reader may wish to take it for granted that the determinant exists for now, skip to Problem C.3.1 to see an application and come back later to read the proofs.

Definition C.3.1 (Determinant)

Let $A = (a_{i,j})$ be a matrix. Denote by $A_{i,j}$ the matrix obtained by removing the ith row and the jth column. We define the *determinant* inductively by $\det[a] = a$ and

$$\det A = a_{1,1} \det A_{1,1} - a_{2,1} A_{2,1} + \ldots + (-1)^n a_{n,1} A_{n,1}.$$

Remark C.3.1

We shall also sometimes denote the determinant of $\begin{bmatrix} a_{1,1} & \cdots & a_{1,n} \\ \vdots & \ddots & \vdots \\ a_{n,1} & \cdots & a_{n,n} \end{bmatrix}$ as

$$\begin{vmatrix} a_{1,1} & \cdots & a_{1,n} \\ \vdots & \ddots & \vdots \\ a_{n,1} & \cdots & a_{n,n} \end{vmatrix}.$$

Here is an example for the determinant of 3×3 matrices.

$$\begin{aligned} \det \begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{bmatrix} &= a_{1,1} \det \begin{bmatrix} a_{2,2} & a_{2,3} \\ a_{3,2} & a_{3,3} \end{bmatrix} - a_{2,1} \det \begin{bmatrix} a_{1,2} & a_{1,3} \\ a_{3,2} & a_{3,3} \end{bmatrix} + a_{3,1} \det \begin{bmatrix} a_{1,2} & a_{1,3} \\ a_{2,2} & a_{2,3} \end{bmatrix} \\ &= a_{1,1}(a_{2,2}a_{3,3} - a_{2,3}a_{3,2}) - a_{2,1}(a_{1,2}a_{3,3} - a_{1,3}a_{3,2}) + a_{3,1}(a_{1,2}a_{2,3} - a_{2,2}a_{1,3}). \end{aligned}$$

We shall get a more explicit formula for the determinant in Theorem C.3.3, but for now we shall use this one. Let's prove that it is linear in each column (we say the determinant is *multilinear* in the columns).

Proposition C.3.1

The determinant is linear in each column, i.e., for any $k \in [n]$, $t \in K$ and $C, C' \in K^n$, we have

$$\det[A_1, \ldots, A_{k-1}, tC + C', A_{k+1}, \ldots, A_n] = t \det[A^1, \ldots, A^{k-1}, C, A^{k+1}, \ldots, A^n] + \det[M^1, \ldots, A^{k-1}, C', A_{k+1}, \ldots, A^n].$$

Proof

This follows easily from our inductive definition. For the sake of convenience we shall write $\det^k_D(M) = \det[M^1, \ldots, M_{k-1}, D, M_{k+1}, \ldots, M^n]$ for any $D \in K^N$ and any M. Let $C = (c_i)$ and $C' = (c'_i)$. First suppose $k = 1$. Then,

$$\det^k_{tC+C'}(A) = \sum_i (tc_i + c'_i) \det A_{i,1} = t \sum_i c_i \det A_{i,1} + \sum_i c'_i \det A_{i,1} = t \det^k_C(A) + \det^k_{C'}(A).$$

Otherwise,

$$\det^k_{tC+C'}(A) = \sum_i a_{i,1} \det^{k-1}_{tC+C'} A_{i,1}$$

which is linear by the induction hypothesis.

■

Thus, the determinant is invariant under addition of two columns of the matrix if and only if the determinant of a matrix with two identical columns is 0. Before showing this however, we prove that the determinant changes by a sign when we exchange two columns. This should satisfy the reader who was disappointed by the lack of symmetry in our definition.

Proposition C.3.2

The determinant of a matrix is multiplied by -1 when exchanging two of its columns (which are distinct).

Proof

Without loss of generality suppose $j > i$. We prove the claim when exchanging two consecutive columns. Iterating this process yields the desired conclusion: indeed if we do the switches

$$i \mapsto i + 1 \mapsto \ldots \mapsto i + (j - i),$$

the kth column goes to the $k - 1$th column for $i + 1 \leq k \leq j$, and then if we do the switches

$$\underbrace{j-1}_{\text{originally } j} \mapsto j - 2 \mapsto \ldots \mapsto j - 1(j - 1 - i)$$

we have exchanged at the end exchanged only the original ith column with the original jth column. In total, we made $(-1)^{(j-i)+(j-1-i)} = -1$ switches of consecutive columns, so the determinant is negated.

To prove that this for consecutive columns, we introduce a similar notation to the one we did before:

$$\det^k_{C,C'}(A) = \det[A^1, \ldots, A^{k-1}, C, C', A^{k+2}, \ldots, A^n].$$

We have

$$\begin{aligned} 0 &= \det^k_{A^k+A^{k+1},A^k+A^{k+1}}(A) \\ &= \det^k_{A^k,A^k}(A) + \det^k_{A^{k+1},A^{k+1}}(A) + \det^k_{A^k,A^{k+1}}(A) + \det^k_{A^{k+1},A^k}(A) \\ &= \det A + \det^k_{A^{k+1},A^k}(A) \end{aligned}$$

as wanted since the first two determinants are zero as the matrices have two equal columns.

■

Proposition C.3.3

The determinant of a matrix with two identical columns is zero.

Proof

By induction again (what else can we do when we defined the determinant inductively?). When $n = 1, 2$ this is obvious. Otherwise, by switching some columns, we may assume the second and third ones are equal by Proposition C.3.2. Then

$$\det A = \sum_i a_{i,1} \det A_{i,1}$$

and all $A_{i,1}$ have two identical columns so habe zero determinant by the induction hypothesis.

■

Exercise C.3.4*. Prove that $\det I_n = 1$.

Exercise C.3.5*. Prove that the determinant of a matrix with a zero column is zero.

Exercise C.3.6. Prove that the determinant of a non-invertible matrix is 0.

With this we can almost prove that the determinant is non-zero if and only if the matrix is invertible. But first, we need to know how to compute a certain kind of determinants: determinants of *upper triangular matrices*, i.e. matrices $M = (a_{i,j})$ such that $a_{i,j} = 0$ for $j > i$

$$\begin{bmatrix} a_{1,1} & 0 & \cdots & 0 \\ a_{2,1} & a_{2,2} & \cdots & 0 \\ \vdots & \vdots & \ddots & 0 \\ a_{n,1} & a_{n,2} & \cdots & a_{n,n} \end{bmatrix}.$$

Here is a sketch of how we are going to proceed to prove that the determinant of an invertible matrix is non-zero, using upper triangular matrices.

The determinant is invariant under *column operations*, i.e. adding a scalar times a column to another column. Indeed,

$$\det^k_{A^k+tA^i}(A) = \det^k_{A^k}(A) + t \det^k_{A^i}(A) = \det A$$

since $\det_{A^i}(A) = 0$ as it has two equal columns. Thus, we shall transform A into an upper triangular matrix using column operations and exchanging columns. The determinant of this matrix will then be $\pm \det A$ so we compute it with the following proposition and as a result we conclude that $\det A \neq 0$. This idea of transforming A into a triangular matrix is also what we did to prove that bases all had the same cardinality in Proposition C.1.2.

Proposition C.3.4

The determinant of an upper triangular matrix $A = (a_{i,j})$ is the product of the elements on the diagonal $a_{1,1} \cdot a_{2,2} \cdot \ldots a_{n,n}$.

Proof

By induction! It's clearly true for $n = 1$ and for $n \geq 2$ we have

$$\det A = a_{1,1} \det A_{1,1} - a_{2,1}A_{2,1} + \ldots + (-1)^n a_{n,1}A_{n,1}.$$

Now notice that $A_{i,1}$ has one row full of zeros (the one correspond to the first row of A) which means $\det A_{i,1} = 0$ by Exercise C.3.5*, i.e. $\det A = a_{1,1} \det A_{1,1} = a_{1,1}a_{2,2} \cdot \ldots \cdot a_{n,n}$ by the induction hypothesis.

■

Theorem C.3.1

Any $n \times n$ matrix A is invertible if and only if its determinant $\det A$ is non-zero.

Proof

As said before, we will transform A into an upper triangular matrix by making column operations and exchanging columns. Since this leaves the space generated by the columns unchanged and changes the determinant by a factor of ± 1, it will suffice to prove that an upper triangular matrix is invertible if and only if its diagonal has no zero element. This is Exercise C.3.7*.

Here is how we do it. We proceed by induction on n ($n = 1$ is trivial as always). If the first row of A is all zero then we can directly apply the induction hypothesis on $A_{1,1}$. Otherwise, suppose that $a_i \neq 0$. By exchanging the ith column with the first one, we can assume $i = 1$.

Now, consider the matrix

$$\left(A^1, A^2 - \frac{a_{1,2}}{a_{1,1}}A^1, A^3 - \frac{a_{1,3}}{a_{1,1}}A^1, \ldots, A^n - \frac{a_{1,n}}{a_{1,1}}A^1\right).$$

It is column equivalent to A and its first row is zero, except for $a_{1,1}$. Now apply the induction hypothesis to this matrix.

■

Exercise C.3.7*. Prove that an upper triangular matrix is invertible if and only if its determinant is non-zero, i.e. if the elements on its diagonal are non-zero.

We are finally ready for some applications. We shall first give a proof that algebraic integers are closed under addition and multiplication using our machinery. In fact we even have the following more general criterion.

Proposition C.3.5

An algebraic number α is an algebraic integer if and only if there is a finitely generated $\mathbb{Z}$-module M such that $\alpha M \subseteq M$. (A module is like a vector space except it can be over any ring, not necessary a field. In this case a $\mathbb{Z}$-module is a space where you can add and subtract elements since multiplication by integers is trivial. Finitely generated means that $M = u_1\mathbb{Z} + \ldots + u_m\mathbb{Z}$ for some $u_1, \ldots, u_m$.)

Proof

If α is an algebraic integer, then we can take $M = \mathbb{Z} + \alpha\mathbb{Z} + \ldots + \alpha^{n-1}\mathbb{Z}$ where n is the degree of α. For the converse, suppose M is a finitely generated $\mathbb{Z}$-module such that $\alpha M \subseteq M$.

Let $u_1, \ldots, u_m$ be a system of generators of M. Write

$$\alpha u_j = \sum_i a_{i,j} u_i$$

with $a_{i,j} \in \mathbb{Z}$. Subtracting αu_j from both sides, we get that the vectors

$$(a_{1,j}, \ldots, a_{j-1,j}, a_{j,j} - \alpha, a_{j+1,j}, \ldots, u_{m,j})$$

are linearly independent over $\mathbb{C}$. Let $A = (a_{i,j})$. The previous remark means that the rows of $A - \alpha I_m$ are linearly dependent, i.e. $A - \alpha I_m$ has determinant zero. But the determinant $\det(A - \alpha I_m)$ is a polynomial in α with integer coefficients since A has integer coordinates. Moreover, from Lemma C.3.1, we see that its leading coefficient is $(-1)^m$ so $\alpha \in \overline{\mathbb{Z}}$ as wanted.

■

Corollary C.3.1

The set of algebraic integers $\overline{\mathbb{Z}}$ is closed under addition and multiplication.

Proof

If α and β are algebraic integers of respective degrees m and n, then

$$M = \mathbb{Z}[\alpha, \beta] := \sum_{0 \leq i \leq m-1, 0 \leq j \leq n-1} \alpha^i \beta^j \mathbb{Z}$$

is a finitely generated $\mathbb{Z}$-module such that $(\alpha + \beta)M \subseteq M$ and $\alpha\beta M \subseteq M$. Thus $\alpha + \beta$ and $\alpha\beta$ also are algebraic integers.

■

Exercise C.3.8. Prove that $\overline{\mathbb{Z}}$ is *integrally closed*, meaning that, if f is a monic polynomial with algebraic integer coefficients, then any of its root is also an algebraic integer. (This is also Exercise 1.5.22†.)

Before presenting more applications, we need two last results. Here is an explicit formula for the determinant which will make our life a lot easier. It can easily be proven by induction. Theorem C.3.3 will determine exactly when those $\varepsilon(\sigma)$ are 1 and when they are -1.

Lemma C.3.1

The determinant of an $n \times n$ matrix $A = (a_{i,j})$ is equal to

$$\det \begin{bmatrix} a_{1,1} & \cdots & a_{1,n} \\ \vdots & \ddots & \vdots \\ a_{n,1} & \cdots & a_{n,n} \end{bmatrix} = \sum_{\sigma \in \mathfrak{S}_n} \varepsilon(\sigma) a_{\sigma(1),1} \cdot \ldots \cdot a_{\sigma(n),n}$$

where the sum is taken over all permutations σ of $[n]$ and $\varepsilon(\sigma) \in \{-1, 1\}$. Moreover, $\sigma(\mathrm{id}) = 1$ (i.e. the coefficient of $a_{1,1} \cdot \ldots \cdot a_{n,n}$ is 1).

Exercise C.3.9*. Prove Lemma C.3.1.

Now, we compute a very important determinant, and then we can move on to applications.

Theorem C.3.2 (Vandermonde Determinant)

Let $x_1, \cdots, x_n \in K$ be elements. We have

$$\det \begin{bmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & \cdots & x_2^{n-1} \\ 1 & x_3 & x_3^2 & \cdots & x_3^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^{n-1} \end{bmatrix} = \prod_{i<j} (x_i - x_j)$$

Proof

Note that this determinant is zero when $x_i = x_j$ for some $i \neq j$ since it then has two equal columns. Now replace x_i by a variable X_i and consider this determinant as a polynomial in $X_1, \ldots, X_n$. The previous observation implies that it is divisible by $X_i - X_j$ for any i, j, i.e. by $\prod_{i<j} X_i - X_j$. In addition, Lemma C.3.1 shows that the degree of the determinant is $\frac{n(n-1)}{2}$ which is the same as $\prod_{i<j} X_i - X_j$ so they are equal up to a multiplicative constant. The same lemma shows that the coefficient of

$$X_1 X_2^2 X_3^3 \cdot \ldots \cdot X_n^n$$

in the determinant is 1, so it is equal to $\prod_{i<j} X_i - X_j$ since it has the same coefficient.

■

From this we deduce a very important corollary.

Corollary C.3.2*

If $x_1, \ldots, x_n$ are distinct numbers, then the vectors

$$(1, x_1, \ldots, x_1^n), \ldots, (1, x_n, \ldots, x_n^n)$$

are linearly independent.

Remark C.3.2

In fact, we didn't need to do all this to prove this corollary. Indeed, the invertibility of the matrix

$$M = \begin{bmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & \cdots & x_2^{n-1} \\ 1 & x_3 & x_3^2 & \cdots & x_3^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^{n-1} \end{bmatrix}$$

when $x_1, \ldots, x_n$ are distinct is exactly what Lagrange interpolation A.1.2 gives us. Indeed, the non-explicit form of the theorem says precisely that the linear map $x \mapsto Mx$ from K^n to itself is surjective. I hope the reader doesn't feel too disheartened by this, there are times where the explicit value of the determinant will be useful to us (in the exercises).

Here is an arithmetic application, which is exactly how we will use the Vandermonde determinant in this book (in the exercises).

Problem C.3.1

Let p is a prime number and $a_1, \ldots, a_m$ integers which are not divisible by p. Suppose $p \mid a_1^k + \ldots + a_m^k$ for $k = 1, \ldots, m$. Prove that $p \mid m$.

Solution

Collect the a_i which are equal modulo p together to get

$$p \mid \sum_{i=1}^{n} c_i b_i^n$$

for some positive integers $\sum_i c_i = m$ and distinct $b_i \in \mathbb{F}_p$. We have a system of equations

$$\begin{cases} c_1 + \ldots + c_n \equiv 0 \\ c_1 b_1 + \ldots + c_n b_n \equiv 0 \\ \ldots\ldots\ldots\ldots\ldots\ldots \\ c_1 b_1^n + \ldots + c_n b_n^n \equiv 0. \end{cases}$$

Since the b_i are distinct modulo p, the vectors $(1, \ldots, b_1^n), \ldots, (1, \ldots, b_n^n)$ are linearly independent in $\mathbb{F}_p$ by Vandermonde. Thus, we must have $c_1 \equiv \ldots \equiv c_n \equiv 0$. Since $m = \sum_i c_i$, we have $p \mid m$ as wanted. ■

As we saw from the first example, matrices are deeply linked with system of linear equations. In hindsight, this is obvious: the system

$$\begin{cases} a_{1,1}x_1 + a_{1,2}x_2 + \ldots + a_{1,n}x_n = b_2 \\ a_{2,1}x_1 + a_{2,2}x_2 + \ldots + a_{1,n}x_n = b_2 \\ \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots \\ a_{n,1}x_1 + a_{n,2}x_2 + \ldots + a_{n,n}x_n = b_n \end{cases}$$

is equivalent to

$$\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \begin{bmatrix} a_{1,1} & \cdots & a_{1,n} \\ \vdots & \ddots & \vdots \\ a_{n,1} & \cdots & a_{n,n} \end{bmatrix} = x_1 \begin{bmatrix} a_{1,1} \\ a_{2,1} \\ \vdots \\ a_{n,1} \end{bmatrix} + x_2 \begin{bmatrix} a_{1,2} \\ a_{2,2} \\ \vdots \\ a_{n,2} \end{bmatrix} + \ldots + x_n \begin{bmatrix} a_{1,n} \\ a_{2,n} \\ \vdots \\ a_{n,n} \end{bmatrix} = \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix}$$

i.e. to $XA = B$ where $X = (x_i)$, $A = (a_{i,j})$ and $B = (b_i)$. In particular, when A is invertible there is a unique solution, so if we have at a some point in a problem we reach such a system of equations and have a trivial solution, then we know what the x_i are since it's the only solution.

As a last application of determinants, we give a different solution to the cows problem C.1.1.

Alternative Solution to the Cows Problem C.1.1

Again, let w_i be weight of the ith cow. Write

$$\sum_{i \in I_k} w_i = \sum_{i \in J_k} w_i$$

where $|I_k| = |J_k| = n$ and $I_k \cup J_k = [2n+1] \setminus \{k\}$ and suppose $w_{2n+1} \in I_k$ for $k \neq 2n+1$.

Consider the system of $2n$ linear equations in $2n$ unknowns

$$\sum_{i\in J_k} w_i - \sum_{i\in I_k, i\neq 2n+1} w_i = -w_{2n+1}$$

for $k = 1, \ldots, 2n$. The determinant of the associated matrix has the form

$$\begin{bmatrix} 0 & \pm 1 & \pm 1 & \cdots & \pm 1 \\ \pm 1 & 0 & \pm 1 & \cdots & \pm 1 \\ \pm 1 & \pm 1 & 0 & \cdots & \pm 1 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \pm 1 & \pm 1 & \pm 1 & \cdots & 0 \end{bmatrix}$$

where there are 0s on the diagonal and a 1 in the (i, j) coordinate if $i \in J_j$ and a -1 otherwise. We wish to show that this determinant is non-zero. Thus, there will be a unique solution to the system, and since $w_1 = \ldots = w_{2n} = w_{2n+1}$ is such a solution it will imply that they are indeed all equal. Modulo 2, the determinant is simply

$$\begin{bmatrix} 0 & 1 & \cdots & 1 \\ 1 & 0 & \cdots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \cdots & 0 \end{bmatrix} = \sum_{\sigma\in\mathfrak{S}_{2n}} \varepsilon(\sigma) a_{1,\sigma(1)} \cdot \ldots \cdot a_{n,\sigma(n)} \equiv \sum_{\sigma\in\mathfrak{S}_{2n}} a_{1,\sigma(1)} \cdot \ldots \cdot a_{2n,\sigma(2n)}$$

where the matrix above is $A = (a_{i,j})$. Now, $a_{1,\sigma(1)} \cdot \ldots \cdot a_{2n,\sigma(2n)}$ is 1 if and only if σ has no fixed point. Thus, this determinant is congruent to the number of *derangements*, i.e. permutations without fixed points. Exercise C.3.10* implies that this number is odd so non-zero, so the original determinant was also odd and in particular non-zero and we are done. ■

Remark C.3.3

One can also compute directly

$$\begin{bmatrix} 0 & 1 & \cdots & 1 \\ 1 & 0 & \cdots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \cdots & 0 \end{bmatrix},$$

as a consequence of, e.g., Exercise C.5.6.

Exercise C.3.10*. Prove that the number of derangements of $[m]$ is

$$\sum_{i=0}^{m} \frac{(-1)^i m!}{i!}$$

and that this number is odd if m is even and even if m is odd.

All right, now let's finish with the determinant.

Definition C.3.2 (Signature)

The *signature* $\varepsilon(\sigma)$ of a permutation σ of $[n]$ is

$$\prod_{1\leq i<j\leq n} \frac{\sigma(i) - \sigma(j)}{i - j}.$$

Note that since σ is a permutation, its signature is in $\{-1, 1\}$. Thus, the signature of a permutation is -1 raised to its number of inversions, i.e. the number of $j > i$ such $\sigma(j) < \sigma(i)$ (an inversion has a contribution of -1 in the signature).

This definition of the signature is in fact not always convenient to work with, so we shall also mention another one. One can see that when we apply a *transposition* to σ, i.e. switch two of its values $\sigma(i)$ and $\sigma(j)$, the signature is multiplied by -1. Since any permutation is a composition of transpositions, the signature is 1 if there are an even number of transpositions and -1 otherwise. In the first case we say the permutation is *even*, and in the second one that it is *odd*.

In particular it does not depend on which transpositions we choose. for example, if one starts with the sequence $(1, \dots, 2m)$ and switches a pair of elements at each step, one will never be able to go back to the original tuple after an odd number of times since the signature will be -1 while the signature of the identity is 1.

As another consequence of this characterisation, we see that the signature is a morphism of groups $\mathfrak{S}_n \to \{-1, 1\}$: indeeed $\varepsilon(\sigma \circ \sigma') = \varepsilon(\sigma) \cdot \varepsilon(\sigma')$ (this is obvious if you consider σ and σ' as a composition of transpositions).

Exercise C.3.11*. Prove that the signature is negated when one exchanges two values of σ (i.e. compose a transposition with σ).

Exercise C.3.12*. Prove that transpositions $\tau_{i,j} : i \leftrightarrow j$ and $k \mapsto k$ for $k \neq i, j$ generate all permutations (through composition).

Remark C.3.4

Proposition C.3.2 now reads as follow: when we apply a transposition to the columns, the determinant gets multiplied by -1. Thus, when we apply a permutation σ to the columns, the determinant gets multiplied by $\varepsilon(\sigma)$. (This is also a direct corollary of the next theorem.)

We get the following refinement of Lemma C.3.1 by induction.

Theorem C.3.3

The determinant of an $n \times n$ matrix $A = (a_{i,j})$ is equal to

$$\det \begin{bmatrix} a_{1,1} & \cdots & a_{1,n} \\ \vdots & \ddots & \vdots \\ a_{n,1} & \cdots & a_{n,n} \end{bmatrix} = \sum_{\sigma \in \mathfrak{S}_n} \varepsilon(\sigma) a_{\sigma(1),1} \cdot \ldots \cdot a_{\sigma(n),n}$$

where the sum is taken over all permutations σ of $[n]$.

Exercise C.3.13*. Prove Theorem C.3.3.

Since permutations are symmetric with respect to the rows and the columns, we get that the determinant is also symmetric with respect to the rows or the columns. In particular, our expansion with respect to one column that we defined the determinant with also holds for rows (and using Proposition C.3.2 it holds for **any** row and any column).

Corollary C.3.3

For any square matrix A, $\det A = \det A^\mathsf{T}$.

Exercise C.3.14*. Prove that $\det A = \det A^\mathsf{T}$ for any square matrix A.

As promised in the beginning of the section, the determinant is the unique solution of a certain functional equation. In fact, this is equation is more or less equivalent to our inductive definition as we shall see. As a consequence, we will see that this implies the multiplicativity of the determinant.

Theorem C.3.4

The determinant of $n \times n$ matrices is the only function D which is multilinear (linear in all columns), zero when two columns are the same, and such that $D(I_n) = 1$.

Proof

We proceed by induction on n. When $n = 1$ it's obvious. For the inductive step, consider the canonical basis of K^n, i.e. the column vectors e_i with a 1 in ith position and zeros everywhere else.

The same proof as Proposition C.3.2 shows that when we exchange two columns, D gets multiplied by -1 (since the only thing we used there was the multilinearity). Note that if $A = (a_{i,j})$ is an $(n-1) \times (n-1)$ matrix, then

$$D_k : A \mapsto D \begin{bmatrix} 0 & a_{1,1} & \cdots & a_{1,n-1} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & a_{n,1} & \cdots & a_{1,n-1} \end{bmatrix}$$

where the first column is E^k, the kth row has only zeros, and the other rows have the matrix A (a bit distorted if $k \neq 1, n$) also satisfies the conditions of the theorem, except possibly the unitary condition. One can check that $D_k(I_{n-1}) = (-1)^{k-1}$ by exchanging some columns (Exercise C.3.15*), thus $D_k(A) = (-1)^k \det A$ by the induction hypothesis. Notice also that, for any $b_1, \ldots, b_{n-1}$

$$D \begin{bmatrix} 0 & a_{1,1} & \cdots & a_{1,n-1} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & b_1 & \cdots & b_{n-1} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & a_{n,1} & \cdots & a_{1,n-1} \end{bmatrix} = D_k(A)$$

since by adding the first column E^k to the other ones we can get rid of the b_i as this doesn't change the determinant.

Finally, using the multilinearity we have

$$D(A) = D\left(\sum_{i=1}^{n} a_{i,1} E^i, A^2, \ldots, A^n\right) = \sum_{i=1}^{n} a_{i,1} D_i(A_{1,i})$$

by the previous remark. Since $D_i(A_{1,i}) = (-1)^{i-1} \det A$, this is the recurrence we originally defined the determinant with so we are done.

■

Exercise C.3.15*. Prove that $D_k(I_{n-1}) = (-1)^{k-1}$.

From this we deduce the following.

Proposition C.3.6 (Multiplicativity of the Determinant)*

The determinant is multiplicative, i.e. for any $n \times n$ matrice A and B we have $\det(AB) = \det(A)\det(B)$.

Proof

Fix B. The function $A \mapsto \det(AB)/\det(B)$ satisfies all conditions of Theorem C.3.4 so is equal to $\det(A)$.

■

Exercise C.3.16. Prove that the determinant is multiplicative by using the explicit formula of Theorem C.3.3.

As an important corollary, we get $\det(A^{-1}) = \det(A)^{-1}$ since

$$\det(A)\det(A^{-1}) = \det(I_n) = 1.$$

This result also lets us define the determinant of a linear map, although we won't be using this here.

Corollary C.3.4

We can define the determinant of a linear map $\varphi : U \to U$ as the determinant of any matrix $M_{\mathcal{B}}^{\mathcal{B}}(\varphi)$ representing φ.

Proof

We need to show that this is well-defined. If M is a matrix representing φ, then the other matrices representing φ have the form NMN^{-1} for an invertible N (see Section C.2). Since the determinant is multiplicative, we have

$$\det(NMN^{-1}) = \det(N)\det(M)\det(N)^{-1} = \det(M)$$

as wanted.

■

Remark C.3.5

With this notion of determinant of a linear map, the norm $N_{L/K}(\alpha)$ is sometimes defined as the determinant of the linear map from L to L defined by $x \mapsto x\alpha$ (see Chapter 6).

Exercise C.3.17. Let L/K be a finite extension. Prove that the determinant of the K-linear map $L \to L$ defined by $x \mapsto x\alpha$ is the norm of α defined in Definition 6.2.3.

Finally, one might be interested in knowing when, say, a matrix with integer coordinates has an inverse with integer coordinates as well. This is achieved by Proposition C.3.6 and the following result, which gives an explicit formula for the inverse of a given matrix.

Proposition C.3.7

The *adjugate* of A, $\operatorname{adj} A := ((-1)^{i+j}\det(A_{j,i}))$ satisfies $A \operatorname{adj} A = (\det A)I_n = \operatorname{adj} AA$.

Remark C.3.6

The transpose of the adjugate, $\operatorname{com} A := ((-1)^{i+j}\det(A_{i,j}))$, is called the *comatrix* of A.

Proof

Let's compute $(b_{i,j}) = A \operatorname{adj} A$. We have $b_{i,j} = \sum_{k=1}^{n} a_{i,k}(-1)^{j+k} \det(A_{j,k})$. When $i = j$ this is the ith row expansion of the determinant so is indeed equal to $\det A$. When $i \neq j$, it is still the expansion of a determinant, but not of A: it is $(-1)^{i+j}$ times the determinant of the matrix obtained by replacing the ith row of A by its jth row. This matrix has two identical rows so its determinant is zero.

Thus we have $b_{i,j} = 0$ if $i \neq j$ and $b_{i,i} = \det A$, i.e. $A \operatorname{adj} A = (\det A) I_n$. The coordinates of $\operatorname{adj} AA$ are treated in a similar fashion, by noting that they are column expansions of certain determinants this time.

■

Remark C.3.7

There is another way to argue that $\operatorname{adj} AA$ is also equal to $(\det A) I_n$ once we have proven $A \operatorname{adj} A$ is. Note that this is a polynomial equation in the coordinates of A, so if it holds sufficiently many times in a fixed infinite field K it must always hold (e.g. by Exercise A.1.7*). Suppose that $\det A$ is non-zero. Then, $\operatorname{adj} A / \det A$ is the inverse of A so it commutes with A as wanted. Finally, $(\operatorname{adj} AA - (\det A) I_n) \det A$ is zero for all $A \subseteq K^{n \times n}$ so must be identically zero, which implies $\operatorname{adj} A = (\det A) I_n$ since the determinant is not the zero polynomial.

Exercise C.3.18*. Prove that $\operatorname{adj} AA = (\det A) I_n$.

In fact, this also gives another proof Theorem C.3.1. This follows from the more general corollary below, which answers our question about invertible matrices with integer coefficients.

Corollary C.3.5*

Let A be a matrix with coefficients in a commutative ring R. A is invertible in R (i.e. A has an inverse with coordinates in R) if and only if $\det A$ is a unit of R.

For instance, a matrix with integer coordinates has an inverse with integer coordinates if and only if its determinant is ± 1.

Proof

If A^{-1} has coordinates in R, then $\det(A) \det(A^{-1}) = 1$ so $\det A$ is a unit. Conversely, if $u \det A = 1$, then $A(u \operatorname{adj}(A)) = I_n$.

■

C.4 Linear Recurrences

In this short section, we derive the formula for linear recurrences using the Vandermonde determinant, which will be used **a lot** in this book.[2]

[2] Mainly in the exercises, though.

Definition C.4.1 (Linear Recurrences)

We say a sequence $(u_n)_{n\in\mathbb{Z}}$ is a *linear recurrence* if there is some $k \geq 1$ and numbers $a_0, \ldots, a_{k-1} \in K$ such that $a_0 \neq 0$ and

$$u_{n+k} = \sum_{i=0}^{k-1} a_i u_{n+i}$$

for all i. The smallest such k is called the *order* of the linear recurrence, and the polynomial $X^k - a_{k-1}X^{k-1} - \ldots - a_1 X - a_0$ is its *characteristic polynomial*. This polynomial is also called the *characteristic polynomial* of the above equation (and the equation is called the equation *associated* with f).

Theorem C.4.1 (Linear Recurrences)

Let K be characteristic zero field and $(u_n)_{n\in\mathbb{Z}}$ be a linear recurrence of elements of K with characteristic polynomials f. Suppose the distinct roots of f are $\alpha_1, \ldots, \alpha_r$ with multiplicity $m_1, \ldots, m_r$. Then, there exist polynomials $f_1, \ldots, f_r$ of degrees less than $m_1, \ldots, m_r$ respectively such that

$$u_n = f_1(n)\alpha_1^n + \ldots + f_r(n)\alpha_r(n)$$

for all $n \in \mathbb{Z}$.

Proof

Let d be the degree of f. Consider the K-vector space of sequences satisfying the equation associated with f (which is indeed a vector space as it is closed under addition). This space has dimension d: indeed for any $(x_0, \ldots, x_{d-1}) \in K^d$ there is a unique sequence solution to the recurrence $(u_n)_n$ such that $u_0 = x_0, \ldots, u_{d-1} = x_{d-1}$. Thus, the dimension of this space is the same as $\dim K^d = d$.

Now we prove that all sequences of the form $u_n = f_1(n)\alpha_1^n + \ldots + f_k(n)\alpha_r^n$ are solutions. Since these sequences also form a K-vector space with generating family given by

$$u_n = n(n-1)\cdot\ldots\cdot(n-(k-1))\alpha_i^n$$

for $k = 0, \ldots, m_i - 1$ and $i = 1, \ldots, r$. Thus we want to have

$$(n+d)(n+d-1)\cdot\ldots\cdot(n+d-(k-1))\alpha_i^{n+j} = \sum_{j=0}^{d-1}(n+j)(n+j-1)\cdot\ldots\cdot(n+j-(k-1))\alpha_i^{n+j}$$

which is equivalent to

$$(X^n f)^{(k)}(\alpha_i) = 0$$

and that's true because α_i is a root of multiplicity $m_i > k$ of $X^n f$.

Finally, to show that all solutions have this form we want to prove that the dimension of the space of solution of this form is the same as the dimension of the space of all solutions, i.e. d. Since our generating family had exactly $m_1 + \ldots + m_r = d$ elements, this is equivalent to it being a basis.

Thus, we want to show it is linearly indepndent. Suppose that a linear combination was zero, i.e.

$$u_n \sum_i f_i(n)\alpha_i^n = 0$$

for all n. We shall prove that $f_i(n) = 0$ for all n and each i, thus implying that $f_i = 0$ since K has characteristic zero. We proceed by induction on $\sum_i \deg f_i$, the base case follows from

the Vandermonde determinant C.3.2. For the induction step, suppose $\deg f_1 \geq 1$ without loss of generality. Consider the sequence

$$v_n = u_{n+1} - \alpha_1 u_n = \sum_i (\alpha_i f_i(n+1) - \alpha_1 f_i(n))\alpha_i^n.$$

Since $\deg(\alpha_i f_i(X+1) - \alpha_1 f_i) \leq \deg f_i$ for $i \geq 1$ and $\deg(\alpha_1(f_i(X+1) - f_i)) \leq \deg f_i - 1$, by the induction hypothesis we have $\alpha_i f_i(X+1) - \alpha_1 f_i = 0$ for all i. This means that they are constant, but we have already treated this case so we are done.

■

Remark C.4.1

This theorem is equivalent to the existence of a (unique) *partial fraction decomposition* of any rational function, meaning that, given a rational function $h = f/g$ with $g = \prod_{i=1}^r (X - \alpha_i)^{m_i}$ and $\deg g > \deg f$, i.e. $\deg h < 0$, there are polynomials f_i of degree at most $m_i - 1$ such that

$$h = \sum_{i=1}^r \frac{f_i}{(X - \alpha_i)^{m_i}}.$$

In fact, if we fix $g = \sum_{i=0}^d a_i X^i$, the rational function with denominator g and negative degree correspond exactly to the generating functions of linear recurrences with characteristic polynomial $\widehat{g} = X^d g(1/X) = \sum_{i=0}^d a_{d-i} X^i$. Indeed, suppose that $(u_n)_{n\geq 0}$ is such a sequence. Then, since $u_{n+d} = -\frac{1}{a_0} \sum_{k=0}^{d-1} a_{d-k} u_{n+k}$, we have

$$\begin{aligned}
h &:= \sum_{n=0}^{\infty} u_n X^n \\
&= \left(\sum_{n=0}^{d-1} u_n X^n\right) + \sum_{n=0}^{\infty} u_{n+d} X^{n+d} \\
&= \left(\sum_{n=0}^{d-1} u_n X^n\right) + \sum_{n=0}^{\infty} -\frac{X^{n+d}}{a_0} \sum_{k=0}^{d-1} a_{d-k} u_{n+k} \\
&= \left(\sum_{n=0}^{d-1} u_n X^n\right) - \frac{1}{a_0} \sum_{k=0}^{d-1} a_{d-k} X^{d-k} \sum_{n=0}^{\infty} u_{n+k} X^{n+k} \\
&= \left(\sum_{n=0}^{d-1} u_n X^n + \sum_{k=0}^{d-1} \frac{a_{d-k}}{a_0} \sum_{n=0}^{k-1} u_n X^{d-k+n}\right) - \frac{1}{a_0} \sum_{k=0}^{d-1} a_{d-k} X^{d-k} \sum_{n=0}^{\infty} u_n X^n \\
&:= f - \frac{(a_0 - g)h}{a_0}
\end{aligned}$$

so $h = f - \frac{(a_0 - g)h}{a_0}$, i.e. $h = \frac{f}{1 - \left(1 - \frac{g}{a_0}\right)} = \frac{a_0 f}{g}$ as wanted. Note that f can be any polynomial of degree less than d since

$$f = u_0 + (u_1 + u_0 a_1/a_0)X + (u_2 + u_1 a_1/a_0 + u_0 a_2/a_0)X^2 + \dots$$

so we can pick the u_i inductively to get any f (equivalently, the matrix of the coefficients of f represented as linear combinations of the u_i is upper-triangular with no zero coordinate on the diagonal so is invertible).

On the other hand, since the roots of $\widehat{g}$ are the $\frac{1}{\alpha_i}$, by our characterisation of linear recurrences, there are polynomials f_i of degree at most $m_i - 1$ such that $u_n = \sum_{i=1}^r f_i(n)(1/\alpha_i)^n$. Hence, we

also have

$$h = \sum_{n=0}^{\infty} X^n \sum_{i=1}^{r} f_i(n)(1/\alpha_i)^n$$
$$= \sum_{i=1}^{r} \sum_{n=0}^{\infty} f_i(n)(X/\alpha_i)^n.$$

Now, note that the f_i are linear combinations of $(X+k)(X+k-1)\cdot\ldots\cdot(X+1)$ so it suffices to prove that the result holds for these polynomials. For this, simply note that differentiating k times

$$\sum_{n=0}^{\infty}(X/\alpha)^n = \frac{1}{1-X/\alpha} = \frac{-\alpha}{X-\alpha}$$

gives

$$\sum_{n=0}^{\infty}(n+k)(n+k-1)\cdot\ldots\cdot(n+1)(X/\alpha)^n = \frac{k!}{(X-\alpha)^{k+1}}$$

as wanted.

Remark C.4.2

This also works for fields of characteristic $p \neq 0$, but one needs the condition that no root has multiplicity $\geq p+1$, otherwise $n \mapsto f_i(n)$ could be identically zero without f_i being zero (e.g. $f_i = n^p - n$). For instance the equation $u_{n+4} = u_n$ has characteristic polynomial $(X-1)^4$ but the space of solutions of the form $\sum_i f_i(n)\alpha_i^n$ has dimension 2 while the space of solutions of $u_{n+4} = u_n$ has dimension 4 so not all solutions have the wanted form.

Exercise C.4.1. Prove that Theorem C.4.1 holds in a field K of characteristic $p \neq 0$ as long as the multiplicities of the roots of the characteristic polynomial are at most p. In particular, for a fixed characteristic equation, it holds for sufficiently large p.

As a corollary, we get the following result, which is not obvious at first sight.

Corollary C.4.1

The product and sum of two linear recurrences are also linear recurrences.

C.5 Exercises

Vector Spaces and Bases

Exercise C.5.1 (Grassmann's Formula)**.** Let U be a vector space and V, W be two finite-dimensional subspaces of U. Prove that

$$\dim(V+W) = \dim V + \dim W - \dim(V \cap W).$$

Exercise C.5.2 (Noether's Lemma)**.** Let U, V, W be finite-dimensional vector spaces, and let $\varphi : U \to V, \psi : V \to W$ be linear maps. Suppose that $\operatorname{im}\varphi \subseteq \ker\psi$. Prove that there exists a linear map τ such that $\psi = \tau \circ \varphi$. Similarly, if $\operatorname{im}\varphi \subseteq \operatorname{im}\psi$, prove that there is a linear map τ such that $\varphi = \psi \circ \tau$.

Exercise C.5.3†**.** Given a vector space V of dimension n, we say a subspace H of V is a *hyperplane* of V if it has dimension $n-1$. Prove that H is a hyperplane of K^n if and only if there are elements $a_1, \ldots, a_n \in K$ not all zero such that

$$H = \{(x_1, \ldots, x_n) \in K^n \mid a_1x_1 + \ldots + a_nx_n = 0\}.$$

Exercise C.5.4. Let E be a vector space with subspaces $E_1, \ldots, E_n$. Suppose that $E_1 \cup \ldots \cup E_n$ is a vector space. Prove that one E_i contains all others.

Exercise C.5.5. Let $z_1, \ldots, z_{n+1} \in \mathbb{C}$ be distinct complex numbers. Prove that $(X - z_1)^n, \ldots, (X + z_{n+1})^n$ is a $\mathbb{C}$-basis of the space of complex polynomials with degree at most n.

Determinants

Exercise C.5.6. Let $a_0, \ldots, a_{n-1}$ be elements of K and ω a primitive nth root of unity. Prove that the *circulant determinant*

$$\begin{bmatrix} a_0 & a_1 & \cdots & a_{n-1} \\ a_{n-1} & a_0 & \cdots & a_{n-2} \\ \vdots & \vdots & \ddots & \vdots \\ a_1 & a_2 & \cdots & a_0 \end{bmatrix}$$

is equal to

$$f(\omega)f(\omega^2) \cdot \ldots \cdot f(\omega^{n-1})$$

where $f = a_0 + \ldots + a_{n-1}X^{n-1}$. Deduce that this determinant is congruent to $a_0 + \ldots + a_{p-1}$ modulo p when $n = p$ is prime and $a_1, \ldots, a_p$ are integers.

Exercise C.5.7 (Cramer's Rule)**.** Consider the system of equations $MV = X$ where M is an $n \times n$ matrix and $V = (v_i)_{i \in [\![1,n]\!]}$ and $X = (x_i)_{i \in [\![1,n]\!]}$ are column vectors. Prove that, for any $k \in [\![1, n]\!]$, v_k is equal to $\det M / \det M_{k,X}$, where $M_{k,X}$ denotes the matrix $[M^1, \ldots, M^{k-1}, X, M^{k+1}, \ldots, M^n]$ obtained from M by replacing the kth column by X.

Exercise C.5.8†**.** Let $(u_n)_{n \geq 0}$ be a sequence of elements of a field K. Suppose that the $(m+1) \times (m+1)$ determinant $\det(u_{n+i+j})_{i,j \in [\![0,m]\!]}$ is 0 for all sufficiently large n. Prove that there is some N such that $(u_n)_{n \geq N}$ is a linear recurrence of order at most m.

Exercise C.5.9†**.** Let $f_1, \ldots, f_n : \mathbb{N} \to \mathbb{C}$ be functions which grow at different rates, i.e.

$$\frac{f_1(m)}{f_2(m)}, \frac{f_2(m)}{f_3(m)}, \ldots, \frac{f_{n-1}(m)}{f_n(m)} \underset{m \to \infty}{\longrightarrow} 0$$

Prove that there exists n integers $m_1, \ldots, m_n$ such that the tuples

$$(f_1(m_1), \ldots, f_n(m_1)), \ldots, (f_1(m_n), \ldots, f_n(m_n))$$

are linearly independent over $\mathbb{C}$.

Exercise C.5.10. Suppose that K is an infinite field and that $A \subseteq K^{n \times n}$ is such that $\det(A + M) = \det M$ for any $M \subseteq K^{n \times n}$. Prove that $A = 0$.

Exercise C.5.11. Let $a_1, \ldots, a_n$ be integers. Prove that

$$\prod_{i<j} \frac{a_i - a_j}{i - j}$$

is an integer by expressing it as the determinant of a matrix with integer coordinates.

Algebraic Combinatorics

Exercise C.5.12†**.** Let $A_1, \ldots, A_{n+1}$ be non-empty subsets of $[n]$. Prove that there exist disjoint subsets I and J of $[n+1]$ such that

$$\bigcup_{i \in I} A_i = \bigcup_{j \in J} A_j.$$

Exercise C.5.13. Let p be a prime number and $a_1, \ldots, a_{p+1}$ real numbers. Suppose that, whenever we remove one of the a_i, we can divide the remaining ones into a certain amount of groups, depending on i, each with the same arithmetic mean (and at least two groups). Prove that $a_1 = \ldots = a_{p+1}$.

Exercise C.5.14. We have n coins of unknown masses and a balance. We are allowed to place some of the coins on one side of the balance and an equal number of coins on the other side. After thus distributing the coins, the balance gives a comparison of the total mass of each side, either by indicating that the two masses are equal or by indicating that a particular side is the more massive of the two. Show that at least $n-1$ such comparisons are required to determine whether all of the coins are of equal mass.

The Characteristic Polynomial and Eigenvalues

Exercise C.5.15 (Characteristic Polynomial)**.** Let K be an algebraically closed field. Let $M \subseteq K^{n\times n}$ be an $n \times n$ matrix. Define its *characteristic polynomial* as $\chi_M = \det(M - XI_n)$. Its roots (counted with multiplicity) are called the *eigenvalues* $\lambda_1, \ldots, \lambda_n \in K$ of M. Prove that $\det M$ is the product of the eigenvalues of M, and that $\operatorname{Tr} M$ is the sum of the eigenvalues. In addition, prove that λ is an eigenvalue of M if and only if there is a non-zero column vector V such that $MV = \lambda V$ (in other words, M acts like a homothety on V). Conclude that, if $f \in \mathbb{C}[X]$ is a polynomial, the eigenvalues of $f(M)$ are $f(\lambda_i)$ (with multiplicity). (We are interpreting $1 \in K$ as I_n for $f(M)$ here, i.e., if $f = X+1$, $f(M)$ is $M + I_n$.) In particular, the eigenvalues of $M + I\alpha$ are $\lambda_1 + \alpha, \ldots, \lambda_n + \alpha$, and the eigenvalues of M^k are $\lambda_1^k, \ldots, \lambda_n^k$.[3]

Exercise C.5.16 (Cayley-Hamilton Theorem)**.** Prove that, for any $n \times n$ matrix M, $\chi_M(M) = 0$ where χ_M is the characteristic polynomial of M and $0 = 0I_n$. Conclude that, if every eigenvalue of M is zero, M is *nilpotent*, i.e. $M^k = 0$ for some k.[4]

Exercise C.5.17. Let $A \subseteq \mathbb{C}^{n\times n}$ be a *Hermitian matrix*, i.e. $A = \overline{A^{\mathsf{T}}}$. Prove that all its eigenvalues are real.

Exercise C.5.18. Let M be a square matrix with integer coordinates and p a prime number. Prove that $\operatorname{Tr} M^p \equiv \operatorname{Tr} M \pmod{p}$.

Exercise C.5.19. Let p be a prime number, and G be a finite (multiplicative) group of $n \times n$ matrices with integer coordinates. Prove that two distinct elements of G stay distinct modulo p. What if the elements of G only have algebraic integer coordinates and p is an algebraic integer with all conjugates greater than 2 in absolute value?

Miscellaneous

Exercise C.5.20 (USA TST 2019)**.** For which integers n does there exist a function $f : \mathbb{Z}/n\mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$ such that

$$f, f + \mathrm{id}, f + 2\mathrm{id}, \ldots, f + m\mathrm{id}$$

are all bijections?

Exercise C.5.21 (Finite Fields Kakeya Conjecture, Zeev Dvir)**.** Let $n \geq 1$ an integer and $\mathbb{F}$ a finite field. We say a set $S \subseteq \mathbb{F}^n$ is a *Kakeya set* if it contains a line in every direction, i.e., for every $y \in \mathbb{F}^n$, there exists an $x \in \mathbb{F}^n$ such that S contains the line $x + y\mathbb{F}$. Prove that any polynomial of degree less than $|\mathbb{F}|$ vanishing on a Kakeya set must be zero. Deduce that there is a constant $c_n > 0$ such that, for any finite field $\mathbb{F}$, any Kakeya set of $\mathbb{F}^n$ has cardinality at least $c_n p^n$.

Exercise C.5.22 (Siegel's Lemma)**.** Let $a = (a_{i,j})$ be an $m \times n$ matrix with integer coordinates. Prove that, if $n > m$, the system

$$\sum_{j=1}^{n} a_{i,j} x_j = 0$$

[3] One of the advantages of the characteristic polynomial is that we are able to use algebraic number theory, or more generally polynomial theory, to deduce linear algebra results, since the eigenvalues say a lot about a matrix (if we combine this with the Cayley-Hamilton theorem). See for instance Exercise C.5.18 and the third solution of Exercise C.5.19.

[4] Note that if, in the definition of χ_M, we replace det by an arbitrary multilinear form in the coordinates of M, such as the *permanent*) $\operatorname{perm}(A) = \sum_{\sigma\in\mathfrak{S}_n} a_{1,\sigma(1)} \cdot \ldots \cdot a_{n,\sigma(n)}$, the result becomes false, so we cannot just say that "$\chi_M(M) = \det(M - MI_n) = \det 0 = 0$" (this "proof" is nonsense because the scalar 0 is not the matrix 0, but the point is that this intuition is fundamentally incorrect).

for $i = 1, \ldots, n$ always has a solution in integers with

$$\max_i |x_i| \leq \left(n \max_{i,j} |a_{i,j}| \right)^{\frac{m}{n-m}}.$$

Exercise C.5.23. Define the *trace* of a matrix as the sum of its diagonal coefficients, and the trace of a linear map as the trace of any matrix representing it. Prove that this doesn't depend on the basis chosen and is thus well-defined. In addition, let L/K be a finite separable extension with embeddings $\sigma_1, \ldots, \sigma_n$ and let $\alpha \in K$. Prove that the trace of the linear map $x \mapsto x\alpha$ is

$$\sum_{i=1}^{n} \sigma_i(\alpha).$$

This function is called the *trace* $\mathrm{Tr}_{L/K}$ *of* L/K.

Exercise C.5.24. How many invertible $n \times n$ matrices are there in $\mathbb{F}_p$? Deduce the number of (additive) subgroups of cardinality p^m that $(\mathbb{Z}/p\mathbb{Z})^n$ has.

Exercise C.5.25†. Let K be a field, and let $S \subseteq K^2$ be a set of points. Prove that there exists a polynomial $f \in K[X, Y]$ of degree at most $\sqrt{2n}$ such that $f(x, y) = 0$ for every $(x, y) \in S$.

Exercise C.5.26†. Given an $m \times n$ matrix M, we define its *row rank* as the maximal number of linearly independent rows of M. Similarly, its *column rank* is the maximal number of linearly independent columns of M. Prove that these two numbers are the same, called the rank of M and denoted $\operatorname{rank} M$.

Exercise C.5.27. Let $A, B \in R^{n \times n}$. Prove that $\operatorname{com}(AB) = \operatorname{com}(A) \operatorname{com}(B)$.

Exercise C.5.28 (Nakayama's Lemma)**.** Let R be a commutative ring, I an *ideal* of a R, i.e. an *R-module* inside R[5], and M a finitely-generated R-module. Suppose that $IM = M$, where IM does not mean the set of products of elements of I and M, but instead the R-module it generates (i.e. the set of linear combinations of products). Prove that there exists an element $r \equiv 1 \pmod{I}$ of R such that $rM = 0$.

[5]See also Proposition C.3.5. A module is like a vector space but the underlying structure is not necessarily a field (in this case it's R).

Appendix C

Linear Algebra

C.1 Vector Spaces

Exercise C.1.1*. Prove Proposition C.1.3.

Solution

Let $u_1, \dots, u_k$ be linearly independent elements of the n-dimensional vector space V. Proceed as follow to complete it into a basis: as long as it does not generate everything, add one element that is not generated (and thus linearly independent with the previous ones). This process must stop since $n+1$ vectors are always linearly dependent by Proposition C.1.2.

For the second part, let $u_1, \dots, u_k$ be a generating family of elements and suppose without loss of generality that $u_1, \dots, u_m$ is a maximal subset of linearly independent elements. This is a basis, since every other u_k can be represented as a linear combination of them (and thus all of V too). Indeed, since $u_1, \dots, u_m, u_k$ are linearly dependent for $k > m$, we have $au_m + \sum_{i=1}^{m} a_i u_i = 0$ for some $a_i \in K$. Since $u_1, \dots, u_m$ are linearly independent, a must be non-zero and we get

$$u_m = -\sum_{i=1}^{m} \frac{a_i}{a} u_i$$

as wanted. ■

C.2 Linear Maps and Matrices

Exercise C.2.1*. Prove that $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ but $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$.

Solution

We have

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 \cdot 1 + 0 \cdot 0 & 1 \cdot 1 + 0 \cdot 0 \\ 0 \cdot 1 + 0 \cdot 0 & 0 \cdot 1 + 0 \cdot 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$$

and

$$\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \cdot 1 + 1 \cdot 0 & 1 \cdot 0 + 1 \cdot 0 \\ 0 \cdot 1 + 0 \cdot 0 & 0 \cdot 0 + 0 \cdot 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}.$$

■

Exercise C.2.2*. Prove that matrix multiplication is *distributive* over matrix addition, i.e. $A(B+C) = AB + AC$ and $(A+B)C = AC + BC$ for any A, B, C of compatible dimensions.

Solution

Let $A = (a_{i,j})$, $B = (b_{i,j})$ and $C = (c_{i,j})$. Then, the (i,j) coordinate of $A(B+C)$ is

$$\sum_k a_{i,k}(b_{k,j} + c_{k,j})$$

which is equal to $\sum_k a_{i,k}b_{k,j} + \sum_k a_{i,k}c_{k,j}$, i.e. the (i,j) coordinate of $AB + AC$. The right-distributivity is completely analogous by symmetry of the left and right multiplication. ■

C.3 Determinants

Exercise C.3.1. Prove that an $m \times n$ matrix can only have a right-inverse if $m < n$, and only a left-inverse if $m > n$. When does such an inverse exist?

Solution

By symmetry, it suffices to consider right-inverses. Suppose that a matrix A has dimensions $m \times n$ and is right-invertible. Consider the surjective linear map from $K^{n\times m}$ to $K^{m\times m}$ defined by $B \mapsto AB$. By surjectivity, $mn \geq m^2$, i.e. $n \geq m$ as wanted.

For the converse, we shall refine a bit our original argument. Note that each column of AB is the sum of linear combinations of the columns of A: if $A^1, \ldots, A^n$ are the columns of A and $B = (b_{i,j})$, then

$$\sum_i b_{i,k}A^i$$

is the kth column of AB. Hence, A has a right-inverse if and only if $n \geq m$ and its columns are linearly independent. ■

Exercise C.3.2*. Prove that $(AB)^\mathsf{T} = B^\mathsf{T}A^\mathsf{T}$ for any $n \times n$ matrices A, B.

Solution

Let $A = (a_{i,j})$ and $B = (b_{i,j})$. The (i,j) coordinate of AB is $\sum_k a_{i,k}b_{k,j}$ so the (i,j) coordinate of its transpose is $\sum_k a_{j,k}b_{k,i} = \sum_k b_{k,i}a_{j,k}$ which is also the (i,j) coordinate of $B^\mathsf{T}A^\mathsf{T}$. ■

Exercise C.3.3. Prove this identity.

Solution

We have

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \begin{bmatrix} ad + b(-c) & a(-b) + ba \\ cd + d(-c) & c(-b) + da \end{bmatrix} = \begin{bmatrix} ad - bc & 0 \\ 0 & ad - bc \end{bmatrix} = (ad - bc)I_2.$$

■

Exercise C.3.4*. Prove that $\det I_n = 1$.

Solution

As always, by induction. From the definition of the determinant, we have $\det I_n = 1 \cdot \det I_{n-1} + 0 \cdot \ldots = \det I_{n-1}$ and $\det I_1 = 1$. ■

Exercise C.3.5*. Prove that the determinant of a matrix with a zero column is zero.

Solution

Suppose that the kth column M^k of M is zero. By Proposition C.3.1 (with $t = 0$), we have

$$\det M = \det_0^k(A) = 0 \det_0^k(M) = 0.$$

■

Exercise C.3.6. Prove that the determinant of a non-invertible matrix is 0.

Solution

Suppose that the columns of M are linearly dependent, i.e. $\sum_i a_i M^i = 0$ for some $a_i \in K$ with $a_k \neq 0$. Then,

$$0 = \det_0^k(M) = \sum_i a_i \det_{M^i}^k(M)$$

by Exercise C.3.5* and Proposition C.3.1. Now, by Proposition C.3.3, all the determinants vanish except the one with $i = k$. Thus, we get $0 = a_k \det M$ which means $\det M = 0$ as wanted. ■

Exercise C.3.7*. Prove that an upper triangular matrix is invertible if and only if its determinant is non-zero, i.e. if the elements on its diagonal are non-zero.

Solution

Let α_i denote the ith element on the diagonal (i.e. the (i, i) coordinate). First, suppose that $\alpha_i \neq 0$ for every i and that

$$\sum_{i=1}^{n} a_i M^i = 0$$

for some $a_i \in K$ not all zero. Consder the least k such that $a_k \neq 0$ and let α denote the (k, k) coordinate of M. The kth coordinate of $\sum_{i=1}^n a_i M^i$ is $\sum_{i=1}^k a_i \alpha_i = a_k \alpha_k$ since the $a_i = 0$ for $i < k$. This means that $a_k = 0$ which contradicts our assumption.

For the converse, let k be such taht $\alpha_k = 0$. Then, the columns $M^k, M^{k+1}, \ldots, M^n$ all have the top k coordinates zero. Thus, we can view them as vectors with $n - k$ coordinates. We have $n - k + 1$ vectors in a space of dimension $n - k$ so they must be linearly dependent. ■

Exercise C.3.8. Prove that $\overline{\mathbb{Z}}$ is *integrally closed*, meaning that, if f is a monic polynomial with algebraic integer coefficients, then any of its root is also an algebraic integer. (This is also Exercise 1.5.22†.)

Solution

Let $f = X^n + \alpha_{n-1}X^{n-1} + \ldots + \alpha_0$ be a monic polynomial with algebraic integer coefficients and let β be one of its roots. Then,

$$M = \mathbb{Z}[\alpha_{n-1}, \ldots, \alpha_0, \beta]$$

is a finitely generated $\mathbb{Z}$-module such that $\beta M \subseteq M$, so β is an algebraic integer. ■

Exercise C.3.9*. Prove Lemma C.3.1.

Solution

By induction:

$$\begin{aligned}\det A &= \sum_{i=1}^{n}(-1)^{i-1}a_{i,1}\det A_{i,1}\\ &= \sum_{i=1}^{n}(-1)^{i-1}a_{i,1}\sum_{\sigma}\varepsilon(\sigma)a_{2,\sigma(2)}\cdot\ldots\cdot a_{n,\sigma(n)}\\ &= \sum_{i=1}^{n}\sum_{\sigma}(-1)^{i-1}\varepsilon(\sigma)a_{i,1}a_{\sigma(2),2}\cdot\ldots\cdot a_{\sigma(n),n}\end{aligned}$$

where the sum is over the bijections $\sigma : [n] \setminus \{1\} \to [n] \setminus \{i\}$. This has the desired form. Finally, note that the sign of $a_{1,1}\cdot\ldots\cdot a_{n,n}$ is $(-1)^{1-1}$ times the sign of $a_{2,2}\cdot\ldots\cdot a_{n,n}$ which is 1 as wanted. ■

Exercise C.3.10*. Prove that the number of derangements of $[m]$ is

$$\sum_{i=0}^{m}\frac{(-1)^i m!}{i!}$$

and that this number is odd if m is even and even if m is odd.

Solution

We shall instead count the number of permutations with at least one fixed point. We use the principle of *inclusion-exclusion*. For a fixed k, there are $(n-1)!$ permutations σ satisfying $\sigma(k) = k$. Thus, we count $n \cdot (n-1)! = n!$ permutations. However, we have counted some permutations twice: the ones which have at least two fixed points. Thus, we remove $\binom{n}{2}\cdot(n-2)!$ from our count (we need to choose two fixed points and then permute the remaining $n-2$ elements arbitrarily). But now, we have removed some permutations too many times: the ones with at least three fixed points, so we must add $\binom{n}{3}\cdot(n-3)!$ to our count, etc. At the end, we get

$$\sum_{k=1}^{n}(-1)^{k-1}\binom{n}{k}(n-k)! = \sum_{k=1}^{n}(-1)^{k-1}\frac{n!}{k!}.$$

If we subtract this from $n!$ (the total number of permutations), we get exactly the wanted formula. ■

Exercise C.3.11*. Prove that the signature is negated when one exchanges two values of σ (i.e. compose a transposition with σ).

Solution

Say that we exchange $\sigma(i)$ with $\sigma(j)$, i.e. we apply the transposition $\tau = \tau_{\sigma(i),\sigma(j)}$. We have

$$\begin{aligned}\varepsilon(\sigma)/\varepsilon(\tau\circ\sigma) &= \frac{\frac{\sigma(i)-\sigma(j)}{j-i}\cdot\prod_{k\neq i}\frac{\sigma(k)-\sigma(j)}{k-i}\cdot\prod_{k\neq j}\frac{\sigma(k)-\sigma(i)}{k-j}}{\frac{\sigma(j)-\sigma(i)}{j-i}\cdot\prod_{k\neq i}\frac{\sigma(k)-\sigma(i)}{k-i}\cdot\prod_{k\neq j}\frac{\sigma(k)-\sigma(j)}{k-j}}\\ &= \frac{\frac{\sigma(i)-\sigma(j)}{j-i}}{\frac{\sigma(j)-\sigma(i)}{j-i}}\\ &= -1\end{aligned}$$

as wanted. ■

Exercise C.3.12*. Prove that transpositions $\tau_{i,j} : i \leftrightarrow j$ and $k \mapsto k$ for $k \neq i, j$ generate all permutations (through composition).

Solution

By induction: we start with $\tau_{n,\sigma^{-1}(n)}$ so that $\sigma \circ \tau_{n,\sigma^{-1}(n)}(n) = n$. Then, ignoring the last element of $\sigma \circ \tau_{1,\sigma^{-1}(1)}$, it is a permutation of $[n-1]$ so a composition of transpositions. We get the wanted result by applying $\tau_{n,\sigma^{-1}(n)} = \tau^{-1}_{n,\sigma^{-1}(n)}$ to both sides. ■

Exercise C.3.13*. Prove Theorem C.3.3.

Solution

We proceed as in Exercise C.3.9*: we have

$$\det(a_{i,j}) = \sum_{i=1}^{n}\sum_{\sigma}(-1)^{i-1}\varepsilon(\sigma)a_{i,1}a_{\sigma(2),2}\cdot\ldots\cdot a_{\sigma(n),n}$$

where the sum is over the bijections $\sigma : [n]\setminus\{1\} \to [n]\setminus\{i\}$. It remains to prove that $\varepsilon(\sigma) = (-1)^{\sigma(1)-1}\varepsilon(\sigma')$ where σ' denotes the bijection $[n]\setminus\{1\} \to [n]\setminus\{\sigma(1)\}$ obtained by forgetting the first element. This is easy: we want to count the number of inversions of σ which are not inversions of σ'. This is exactly the number of inversions $(k, 1)$ since 1 is the only difference bewteen σ and σ'. Since $k > 1$ for each $k \neq 1$, the number of such inversions is the number of $\sigma(k) < \sigma(1)$, i.e. $\sigma(1) - 1$ as wanted. ■

Exercise C.3.14*. Prove that $\det A = \det A^{\mathsf{T}}$ for any square matrix A.

Solution

Our formula from Theorem C.3.3 is symmetric in rows and columns:

$$\begin{aligned}\det A &= \sum_{\sigma\in\mathfrak{S}_n} \varepsilon(\sigma) a_{\sigma(1),1}\cdot\ldots\cdot a_{\sigma(n),n}\\ &= \sum_{\sigma\in\mathfrak{S}_n} \varepsilon(\sigma^{-1}) a_{1,\sigma^{-1}(1)}\cdot\ldots\cdot a_{n,\sigma^{-1}(n)}\\ &= \sum_{\sigma\in\mathfrak{S}_n} \varepsilon(\sigma) a_{1,\sigma(1)}\cdot\ldots\cdot a_{n,\sigma(n)}\\ &= \det A^{\mathsf{T}}.\end{aligned}$$

■

Exercise C.3.15*. Prove that $D_k(I_{n-1}) = (-1)^{k-1}$.

Solution

Consider the $n\times n$ matrix I we defined D_k with and substitute $A = I_{n-1}$. We will exchange $k-1$ columns to transform it into I_n, thus getting a determinant of $(-1)^{k-1}\det I_n = (-1)^{k-1}$ as wanted.

Note that I is already almost equal to I_n: its first column should be kth and the 1 to k ones should be shifted to the left. Here is how we do this shift using transpositions.

We first exchange the first column of the with the second one so that $a_{1,1}=1$ becomes the $(1,1)$ coordinate of I, then we exchange the (new) second column with the third one so that $a_{2,2}=1$ becomes the $(2,2)$ coordinate of I, etc., until we exchange the $k-1$th column with the kth one so that $a_{k,k}=1$ becomes the (k,k) coordinate of I. Thus, the $k-1$th column, which was originally the first one, becomes the kth one as wanted. ■

Exercise C.3.16. Prove that the determinant is multiplicative by using the explicit formula of Theorem C.3.3.

Solution

Write $C = AB$, so that

$$C^k = b_{1,k}A^1 + \ldots + b_{n,k}A^n.$$

Then, by multilinearity of the determinant,

$$\begin{aligned}\det C &= \det(b_{1,1}A^1+\ldots+b_{n,1}A^n,\ldots,b_{1,n}A^1+\ldots+b_{n,n}A^n)\\ &= \sum_{\sigma\in\mathfrak{S}_n} b_{\sigma(1),1}\cdot\ldots\cdot b_{\sigma(n),n}\det(A^{\sigma(1)},\ldots,A^{\sigma(n)})\\ &= \sum_{\sigma\in\mathfrak{S}_n} \sigma(\varepsilon) b_{\sigma(1),1}\cdot\ldots\cdot b_{\sigma(n),n}\det(A^1,\ldots,A^n)\\ &= \det B\det A\end{aligned}$$

where the second-to-last equality comes from Remark C.3.4. ■

Exercise C.3.17. Let L/K be a finite extension. Prove that the determinant of the K-linear map $L\to L$ defined by $x\mapsto x\alpha$ is the norm of α defined in Definition 6.2.3.

Solution

We first treat the case where $L = K(\alpha)$. Consider the basis $1, \alpha, \ldots, \alpha^{n-1}$ of L and let $f = X^n + a_{n-1}X^{n-1} + \ldots + a_0$ be the minimal polynomial of α. The matrix corresponding to $x \mapsto x\alpha$ in this basis is

$$\begin{bmatrix} 0 & 0 & 0 & \cdots & -a_0 \\ 1 & 0 & 0 & \cdots & -a_1 \\ 0 & 1 & 0 & \cdots & -a_2 \\ 0 & 0 & 1 & \cdots & -a_3 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & -a_{n-1} \end{bmatrix}.$$

Using Theorem C.3.3, we see that this is $-\varepsilon(\sigma)a_0$, where σ is the cycle $(1, 2, \ldots, n)$. Hence, we need to prove that $\varepsilon(\sigma) = (-1)^{n-1}$ since the product of the conjugates of α is $(-1)^n a_0$ by Vieta's formulas. This follows from the fact that

$$\sigma = (1, n) \cdot \ldots \cdot (1, 3)(1, 2)$$

is a product of $n - 1$ transpositions.

Now here is what we can do for the general case. Note that the norm is multiplicative. Indeed, for any linear maps φ and ψ,

$$\det \varphi \cdot \det \psi = \det \varphi \circ \psi$$

since the determinant is multiplicative. Since the composite of $x \mapsto \alpha x$ and $x \mapsto \beta x$ is $x \mapsto \alpha\beta x$, the norm of $\alpha\beta$ is the norm of α times the norm of β. Then, pick a primitive element γ of L/K such that α/γ is also a primitive element. We can do this since

$$\sigma(\alpha)/\sigma(\gamma) = \alpha/\gamma \iff \sigma(\gamma) = \gamma \cdot \frac{\sigma(\alpha)}{\alpha}$$

is false for $\gamma = a\delta + b$ for well-chosen $a, b \in K$ and a fixed primitive element δ. Thus, from the first observation, we get

$$\begin{aligned} N_{L/K}(\alpha) &= N_{L/K}(\gamma) N_{L/K}(\alpha/\gamma) \\ &= \prod_{\sigma \in \mathrm{Emb}_K(L)} \sigma(\gamma) \prod_{\sigma \in \mathrm{Emb}_K(L)} \sigma(\alpha/\gamma) \\ &= \prod_{\sigma \in \mathrm{Emb}_K(L)} \sigma(\alpha) \end{aligned}$$

as wanted. ■

Exercise C.3.18*. Prove that $\operatorname{adj} AA = (\det A) I_n$.

Solution

Set $(b_{i,j}) = \operatorname{adj} AA$. This time, we have

$$b_{i,j} = \sum_{k=1}^{n} (-1)^{i+k} \det(A_{k,i}) a_{k,j}.$$

When $i = j$ this is the column expansion of the determinant of A which is $\det A$, and when $i \neq j$ this is $(-1)^{i+j}$ times the determinant of the matrix obtained by replacing the ith column of A by its jth column. This matrix has two identical columns so its determinant is zero as wanted. ■

C.4 Linear Recurrences

Exercise C.4.1. Prove that Theorem C.4.1 holds in a field K of characteristic $p \neq 0$ as long as the multiplicities of the roots of the characteristic polynomial are at most p. In particular, for a fixed characteristic equation, it holds for sufficiently large p.

Solution

The only thing to check is that if $f_i(n) = 0$ for all $n \in \mathbb{Z}$ and f_i has less than the multiplicity of α_i so less than p then $f_i = 0$. This is true because $\mathbb{Z}$ reduces to p elements in a field of characteristic $p \neq 0$ so f_i has p roots which is more than its degree and is thus zero. ■

C.5 Exercises

Vector Spaces and Bases

Exercise C.5.1 (Grassmann's Formula)**.** Let U be a vector space and V, W be two finite-dimensional subspaces of U. Prove that

$$\dim(V + W) = \dim V + \dim W - \dim(V \cap W).$$

Solution

Let $u_1, \ldots, u_k$ be a basis of $V \cap W$. Complete it to a basis $u_1, \ldots, u_k, v_1, \ldots, v_m$ of V and a basis $u_1, \ldots, u_k, w_1, \ldots, w_n$ of W. We claim that

$$u_1, \ldots, u_k, v_1, \ldots, v_m, w_1, \ldots, w_n$$

is a basis of $V + W$. It clearly spans all of $V + W$ so it remains to check that it's linearly independent. If

$$\sum_{i=1}^{k} a_i u_i + \sum_{i=1}^{m} b_i v_i + \sum_{i=1}^{m} c_i w_i = 0,$$

then $\sum_{i=1}^{m} b_i v_i$ is both in V and W so is in $V \cap W$. This means that it's a linear combination of the u_i, but by construction this implies $b_1 = \ldots = b_m = 0$ since the u_i and v_i are linearly independent (they form a basis of V together). By symmetry, $c_1 = \ldots = c_n = 0$. Finally, this forces $a_1 = \ldots = a_k = 0$ too.

We conclude that

$$\dim(V + W) = k + m + n = (k + m) + (k + n) - k = \dim V + \dim W - \dim V \cap W.$$

■

Exercise C.5.3† **.** Given a vector space V of dimension n, we say a subspace H of V is a *hyperplane* of V if it has dimension $n - 1$. Prove that H is a hyperplane of K^n if and only if there are elements $a_1, \ldots, a_n \in K$ not all zero such that

$$H = \{(x_1, \ldots, x_n) \in K^n \mid a_1 x_1 + \ldots + a_n x_n = 0\}.$$

Solution

Clearly, if H is defined as the zero set of $a_1 X_1 + \ldots + a_n X_n$ then H has dimension $n - 1$ since,

assuming without loss of generality that $a_n \neq 0$, we get a bijective map $K^{n-1} \to H$ given by

$$(x_1, \ldots, x_{n-1}) \mapsto \left(x_1, \ldots, x_{n-1}, -\frac{a_1 x_1 + \ldots + a_n x_n}{a_1}\right).$$

For the converse, pick a linear map φ mapping H to 0 without being identically 0. Then,

$$(x_1, \ldots, x_n) = x \in H \iff \varphi(x) = 0 \iff \sum_{i=1}^{n} x_i \varphi(e_i) = 0$$

where e_i is the canonical basis of K^n: 0 everywhere except in its ith coordinate where there is a 1. ■

Determinants

Exercise C.5.6. Let $a_0, \ldots, a_{n-1}$ be elements of K and ω a primitive nth root of unity. Prove that the *circulant determinant*

$$\begin{bmatrix} a_0 & a_1 & \cdots & a_{n-1} \\ a_{n-1} & a_0 & \cdots & a_{n-2} \\ \vdots & \vdots & \ddots & \vdots \\ a_1 & a_2 & \cdots & a_0 \end{bmatrix}$$

is equal to

$$f(\omega) f(\omega^2) \cdot \ldots \cdot f(\omega^{n-1})$$

where $f = a_0 + \ldots + a_{n-1} X^{n-1}$. Deduce that this determinant is congruent to $a_0 + \ldots + a_{p-1}$ modulo p when $n = p$ is prime and $a_1, \ldots, a_p$ are integers.

Solution

Let A be the circulant matrix. We shall imitate the proof of Theorem C.3.2: we need to prove that $\det A$ is zero when $a_0 = -(a_1\omega + \ldots + a_{n-1}\omega^{n-1})$ for any root of unity ω. This shows that the product $f(\omega)$ divides the determinant as a polynomial in a_0. Then, by looking at the coefficient of a_0^n, we can conclude that they are in fact equal. To show that the determinant vanishes for these a_0, note that the following linear combination of the columns is zero

$$\sum_{i=1}^{n} \omega^i A^i = 0.$$

Alternatively, one could note that $A = f(J)$, where is the circulant matrix with $a_1 = 1$ and $a_0 = a_2 = a_3 = \ldots = a_{n-1} = 0$. We can check that all nth roots of unity are eigenvalues of J. To finish, Exercise C.5.15 implies that the eigenvalues of $f(J)$ are the $f(\omega)$, and the determinant is their product as wanted. ■

Exercise C.5.7 (Cramer's Rule)**.** Consider the system of equations $MV = X$ where M is an $n \times n$ matrix and $V = (v_i)_{i \in [\![1,n]\!]}$ and $X = (x_i)_{i \in [\![1,n]\!]}$ are column vectors. Prove that, for any $k \in [\![1, n]\!]$, v_k is equal to $\det M / \det M_{k,X}$, where $M_{k,X}$ denotes the matrix $[M^1, \ldots, M^{k-1}, X, M^{k+1}, \ldots, M^n]$ obtained from M by replacing the kth column by X.

Solution

Note that this formula is linear in XX since the determinant is multilinear. Since the kth coordinate of the formula $M^{-1}V$ is also linear in X, we just need to check that both formulas

agree on a basis of K^n. This is easy: when $X = M^k$ we get $v_k = 1$ and $v_i = 0$ for $i \neq k$ which is indeed the solution to $MV = M^k$. Since these form a basis of K^n as M is invertible, we are done. ■

Exercise C.5.8†. Let $(u_n)_{n\geq 0}$ be a sequence of elements of a field K. Suppose that the $(m+1)\times(m+1)$ determinant $\det(u_{n+i+j})_{i,j\in[\![0,m]\!]}$ is 0 for all sufficiently large n. Prove that there is some N such that $(u_n)_{n\geq N}$ is a linear recurrence of order at most m.

Solution

We proceed by induction on m (it's trivial when $m = 1$). More precisely, we prove that, under the assumptions of the problem, if the $(m-1)\times(m-1)$ determinant $\det(u_{n+i+j})_{i,j\in[\![0,m-1]\!]}$ vanishes for one value of n, then it vanishes for all the next ones which means that there exists some N for which $(u_n)_{n\geq N}$ is a linear recurrence of order at most $m-1 \leq m$. If it doesn't vanish for $n \geq N$, then $(u_{n+i+j})_{i,j\in[\![0,m]\!]}$ has rank $m-1$ by definition of the rank (see Exercise C.5.26†), and, more precisely, its first m rows as well as its first last rows are linearly independent and generate the same hyperplane H. Notice that the last m rows of $(u_{n+i+j})_{i,j\in[\![0,m]\!]}$ are the first m rows of $(u_{n+1+i+j})_{i,j\in[\![0,m]\!]}$ so this hyperplane H is always the same. Finally, with Exercise C.5.3†, we conclude that

$$a_0u_n + a_1u_{n+1} + \ldots + a_{n+m}u_{n+m} = 0$$

for all $n \geq N$, i.e. $(u_n)_{n\geq N}$ is a linear recurrence of order at most m as claimed.

It remains to prove that, if $\det(u_{n+i+j})_{i,j\in[\![0,m-1]\!]} = 0$, then $\det(u_{n+1+i+j})_{i,j\in[\![0,m-1]\!]} = 0$ as well. Hence, suppose that the first determinant is 0, i.e. that there is a linear dependence between the rows. If this dependence does not involve the first row, then it also creates a linear dependence in the rows of the second matrix which implies that its determinant is 0 as wanted. Otherwise, the first row is a linear combination of the $m-1$ next ones. This implies that the first row of $(u_{n+i+j})_{i,j\in[\![0,m]\!]}$ is a linear combination of the $m-1$ next ones as well as a vector of the form $(0,\ldots,0,a)$ for some $a \in K$. Then, by performing row operations, we find

$$0 = (u_{n+i+j})_{i,j\in[\![0,m]\!]} = \pm\begin{vmatrix} 0 & \cdots & 0 & a \\ u_{n+1} & \cdots & u_{n+m} & u_{n+m+1} \\ \vdots & \vdots & \ddots & \vdots \\ u_{n+m} & \cdots & u_{n+2m-1} & u_{n+2m} \end{vmatrix} = \pm a \det\det(u_{n+1+i+j})_{i,j\in[\![0,m-1]\!]}$$

by expanding with respect to the first row. We are done. ■

Exercise C.5.9†. Let $f_1,\ldots,f_n : \mathbb{N} \to \mathbb{C}$ be functions which grow at different rates, i.e.

$$\frac{f_1(m)}{f_2(m)}, \frac{f_2(m)}{f_3(m)}, \ldots, \frac{f_{n-1}(m)}{f_n(m)} \underset{m\to\infty}{\longrightarrow} 0$$

Prove that there exists n integers $m_1,\ldots,m_n$ such that the tuples

$$(f_1(m_1),\ldots,f_n(m_1)),\ldots,(f_1(m_n),\ldots,f_n(m_n))$$

are linearly independent over $\mathbb{C}$.

Solution

We proceed by induction on n. It is of course trivial when $n = 1$. Fix $m_1,\ldots,m_{n-1}$ such that

$$(f_1(m_1),\ldots,f_{n-1}(m_1)),\ldots,(f_1(m_{n-1}),\ldots,f_{n-1}(m_{n-1}))$$

are linearly independent, i.e. such that the determinant

$$C = \begin{bmatrix} f_1(m_1) & \cdots & f_{n-1}(m_1) \\ \vdots & \ddots & \vdots \\ f_1(m_{n-1}) & \cdots & f_{n-1}(m_{n-1}) \end{bmatrix}$$

is non-zero. We wish to show that there is some m_n such that

$$\begin{bmatrix} f_1(m_1) & \cdots & f_n(m_1) \\ \vdots & \ddots & \vdots \\ f_1(m_{n-1}) & \cdots & f_n(m_n) \end{bmatrix}$$

is non-zero. Expand it with respect to the last column to get

$$\sum_{i=1}^{n} C_i f_i(m_n)$$

where C_i are constants and $C_n = C$. Since f_n dominates all other f_i and C_n is non-zero by construction, this is non-zero for sufficiently large m as wanted. ■

Algebraic Combinatorics

Exercise C.5.12†. Let $A_1, \ldots, A_{n+1}$ be non-empty subsets of $[n]$. Prove that there exist disjoint subsets I and J of $[n+1]$ such that

$$\bigcup_{i \in I} A_i = \bigcup_{j \in J} A_j.$$

Solution

Identity each subset $A_i \subseteq [n]$ with the vector $v_i \in \mathbb{R}^n$ whose ith coordinate is 1 if $i \in S$ and 0 otherwise. This makes the set of subsets of $[n]$ into a subset of a $\mathbb{R}$-vector space of dimension n. (It's not a vector space itself, though it would be if we chose $\mathbb{F}_2$ instead of $\mathbb{R}$. $\mathbb{F}_2$ is usually very useful in algebraic combinatorics but doesn't work here.) We have $n+1$ vectors in a space of dimension n so they must be linearly dependent, say

$$\sum_{i=1}^{n+1} c_i v_i = 0.$$

Now consider the set I of indices of positive c_i, and the set J of indices of negative c_i. We have

$$\sum_{i \in I} |c_i| v_i = \sum_{j \in J} |c_j| v_j$$

which gives us $\bigcup_{i \in I} A_i = \bigcup_{j \in J} A_j$ as wanted. In addition, I and J are disjoint by construction. ■

The Characteristic Polynomial and Eigenvalues

Exercise C.5.15 (Characteristic Polynomial)**.** Let K be an algebraically closed field. Let $M \subseteq K^{n \times n}$ be an $n \times n$ matrix. Define its *characteristic polynomial* as $\chi_M = \det(M - XI_n)$. Its roots (counted with multiplicity) are called the *eigenvalues* $\lambda_1, \ldots, \lambda_n \in K$ of M. Prove that $\det M$ is the product of the eigenvalues of M, and that $\operatorname{Tr} M$ is the sum of the eigenvalues. In addition, prove that λ is an eigenvalue of M if and only if there is a non-zero column vector V such that $MV = \lambda V$ (in other words, M acts like a homothety on V). Conclude that, if $f \in \mathbb{C}[X]$ is a polynomial, the eigenvalues of

$f(M)$ are $f(\lambda_i)$ (with multiplicity). (We are interpreting $1 \in K$ as I_n for $f(M)$ here, i.e., if $f = X+1$, $f(M)$ is $M + I_n$.) In particular, the eigenvalues of $M + I\alpha$ are $\lambda_1 + \alpha, \dots, \lambda_n + \alpha$, and the eigenvalues of M^k are $\lambda_1^k, \dots, \lambda_n^k$.[1]

Solution

The first part follow simply from expanding the determinant and using Vieta's formulas. For the second one, there is a non-zero vector V such that $MV = \lambda V \iff (M - \lambda I_n)V$ if and only if $\ker(M - \lambda I_n)$ is non-trivial, which is equivalent to $\det(M - \lambda I_n) = 0$. Now, note that $MV = \lambda V$ gives $M^k V = \lambda^k V$ and thus, by taking linear combinations, $f(M)V = f(\lambda)V$. This shows that the $f(\lambda)$ are eigenvalues of M. To account for the multiplicity, note that we have established the result when χ_M has simple roots, and that the general results follows by density, analytical or algebraic. We present the proof by algebraic density (more technically, *Zariski density*) since it works over any field, which is similar to Remark C.3.7.

Note that the equality $\chi_{f(M)} = \prod_{k=1}^n f(\lambda_i) - X$ is a polynomial equality in the coefficients of f and the coordinates of M by the fundamental theorem of symmetric polynomials. Let Δ_f by the discriminant of $\prod_{k=1}^n f(\lambda_i) - X$, i.e. $\pm \prod_{i \neq j} f(\lambda_i) - f(\lambda_j)$ which is again polynomial in the coefficients of f and the coordinates of M. We have shown that

$$(\chi_{f(M)} - \prod_{k=1}^{n} f(\lambda_i) - X)\Delta_f$$

always takes the value zero. Hence, it must be the zero polynomial. If we show that Δ_f is non-zero, we are hence done. Choosing $f = X$, this amounts to finding a matrix with distinct eigenvalues. This is not very hard: we can fix all coordinates of M except one and let it vary. The determinant then varies affinely in this coordinates, say it is $ua + v$, where a is the varying coordinate. By induction, we can choose u to have simple roots. A multiple root of $ua + v$ would also be a root of u', but this is impossible for large a unless this root is a common root of u and v, which must thus have multiplicity one: this is a contradiction. (Alternatively, we can consider the matrix J from Exercise C.5.6.) ■

Exercise C.5.16 (Cayley-Hamilton Theorem)**.** Prove that, for any $n \times n$ matrix M, $\chi_M(M) = 0$ where χ_M is the characteristic polynomial of M and $0 = 0I_n$. Conclude that, if every eigenvalue of M is zero, M is *nilpotent*, i.e. $M^k = 0$ for some k.[2]

Solution

Using Proposition C.3.7, we get

$$(M - XI_n)\operatorname{adj}(M - XI_n) = \operatorname{adj}(M - XI_n)(M - XI_n) = \chi_M I_n. \qquad (*)$$

We wish to substitute M for X in $M - XI_n$, but we can only do that if M commutes with the coefficients of $\operatorname{adj}(M - XI_n)$ (which are matrices). Note that this is the case since

$$XI_n \operatorname{adj}(M - XI_n) = \operatorname{adj}(M - XI_n)XI_n$$

(X is a formal variable so commutes with everything, and I_n does as well) so $M \operatorname{adj}(M - XI_n) = \operatorname{adj}(M - XI_n)M$ by $(*)$. The second part is obvious: if all eigenvalues of M are zero, then $\chi_M = \pm X^n$ so $M^n = 0$. ■

[1] One of the advantages of the characteristic polynomial is that we are able to use algebraic number theory, or more generally polynomial theory, to deduce linear algebra results, since the eigenvalues say a lot about a matrix (if we combine this with the Cayley-Hamilton theorem). See for instance Exercise C.5.18 and the third solution of Exercise C.5.19.

[2] Note that if, in the definition of χ_M, we replace det by an arbitrary multilinear form in the coordinates of M, such as the *permanent*) $\operatorname{perm}(A) = \sum_{\sigma \in \mathfrak{S}_n} a_{1,\sigma(1)} \cdot \ldots \cdot a_{n,\sigma(n)}$, the result becomes false, so we cannot just say that "$\chi_M(M) = \det(M - MI_n) = \det 0 = 0$" (this "proof" is nonsense because the scalar 0 is not the matrix 0, but the point is that this intuition is fundamentally incorrect).

Exercise C.5.19. Let p be a prime number, and G be a finite (multiplicative) group of $n \times n$ matrices with integer coordinates. Prove that two distinct elements of G stay distinct modulo p. What if the elements of G only have algebraic integer coordinates and p is an algebraic integer with all conjugates greater than 2 in absolute value?

Solution

We will present three solutions, in increasing order of non-elementariness, and of generality. Suppose that the reduction modulo p (group) morphism φ is not injective, i.e. has non-trivial kernel (see Exercise A.2.13* – this is really trivial, I'm only using this language so that you familiarise with it), say $I := I_n \neq A \subseteq\in \ker \varphi$, i.e. $A = I + pB$ for some non-zero B. Since G is a finite group, we have $A^m = I$ for $m = |G|$ by Lagrange's theorem Exercise A.3.17†. We will show that this is impossible (if fact it is equivalent to the problem: if it were possible, the group generated by M would be a counterexample). (Note that all solutions will use the assumption that $p \neq 2$ somewhere, and for a good reason: $-I \equiv I \pmod 2$.)

Here is the first solution, which works only for the first part of the problem. Let k be the greatest integer such that B/p^k has integer coordinates. Suppose first that $p \nmid m$. Then, modulo p^{k+2}, using the binomial expansion (which we can use since I and pB commute), we have

$$(I + pB)^m \equiv I + mpB \not\equiv I \pmod{p^{k+2}}$$

which is a contradiction. We will now replace B by C such that $(I + pB)^p = I + pC$. That, way, m gets replaced by m/p, and by iterating this process we will eventually reach a $p \nmid m$ which is a contradiction. However, we need to prove that $C \neq 0$ too. Thus, suppose that $(I + pB)^p = I$. We then have

$$(I + pB)^m \equiv I + mpB + \frac{m(m-1)}{2}p^2B^2 equiv I + p^2B \not\equiv I \pmod{p^{k+3}}$$

since p is odd, which is also a contradiction.

We now present the second solution. Let $|M|$ denote the maximum of the absolute value of the coordinates of a matrix M. Since $A^m = I$ for some m, $|A^k|$ is bounded when k varies, say by C. We have

$$|(pB)^r = |(I - A)^r| \leq \sum_{k=0}^{r} \binom{r}{k} |A^r| \leq C2^r.$$

However, the left hand side is divisible by p^n so is at least p^n unless it is 0. Since $p > 2$, by taking r sufficiently large, we get $|B^r| = 0$, i.e. $B^r = 0$: B is nilpotent. When p is only an algebraic integer with all conjugates greater than 2, the same argument works: the coordinates of B^r/p^r are algebraic integers with absolute value less than $C2^r/p^r$ so less than 1 for large r. However, the same goes for their conjugates by assumption. Since the only algebraic integer whose conjugates are all strictly in the unit disk is 0 (by looking at the constant coefficient of its minimal polynomial), we get $B^r = 0$ for large r as wanted.

Consider k such that $B^k \neq 0$ but $B^{k+1} = 0$. Since $B \neq 0$, we have $k \geq 1$. By expanding $(I + pB)^m = I$, we get

$$pmB + p^2\frac{m(m-1)}{2}B^2 + \ldots = 0.$$

Finally, by multiplying this equation by B^{k-1}, we get $pmB^k = 0$ which implies $m = 0$ and is a contradiction.

Finally, the third solution uses more advanced linear algebra. Let β be an eigenvalue of B. Then, $\alpha = 1 + p\beta$ is an eigenvalue of A which is congruent to 1 modulo p. Further, since A^m, we have $\alpha^m = 1$ so it is a root of unity. This implies that $\beta = \frac{\alpha-1}{p}$ has module less than 1 since $p > 2$. This is also true for all its conjugates. Thus, the constant coefficient of its minimal polynomial must be 0, i.e. $\beta = 0$. Hence, all eigenvalues of B are zero, which implies that B is nilpotent by Exercise C.5.16. We finish as in the previous solution. ■

Miscellaneous

Exercise C.5.20 (USA TST 2019)**.** For which integers n does there exist a function $f : \mathbb{Z}/n\mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$ such that

$$f, f + \mathrm{id}, f + 2\mathrm{id}, \ldots, f + m\mathrm{id}$$

are all bijections?

Solution

There exists such a function if and only if all prime factors of n are greater than $m+1$. In that case, it is clear that $f = \mathrm{id}$ works. Now suppose that f has a prime factor $p \leq m+1$. Pick p to be minimal, and suppose without loss of generality that $m = p-1$. For $1 \leq k \leq m$, since f and $f + k\mathrm{id}$ are both bijections, we have

$$\sum_{x=1}^{n} g(x)^m \equiv \sum_{x=1}^{n} (g(x) + kx)^m = \sum_{i=0}^{m} \binom{m}{i} k^{m-i} \sum_{x=1}^{n} g(x)^i x^{m-i},$$

i.e.

$$\sum_{i=0}^{m-1} \binom{m}{i} k^{m-i} \sum_{x=1}^{n} g(x)^i x^{m-i} \equiv 0.$$

Thus, we have a linear system in the k^{m-i} with solution $x_i = \binom{m}{i} \sum_{x=1}^{n} g(x)^i x^{m-i}$. By Vandermonde, the determinant of the matrix $M = (k^{m-i})_{k,i}$ is

$$\prod_{1 \leq i < j \leq m} i - j$$

which is invertible modulo n since p is the smallest prime factor of n by assumption and $m = p-1$. Thus, by Exercise C.3.18*, M is invertible modulo n which implies that our system has exactly one solution. Since $x_0 \equiv \ldots \equiv x_{m-1} = 0$ is of course the trivial solution, this must in fact be the case. In particular,

$$x_0 = \sum_{x=1}^{n} x^{p-1}$$

is zero. We will prove that this is impossible. If $p = 2$, this sum is $\frac{n}{n-1}2$ so $n \mid \frac{n(n-1)}{2}$ which implies that n is odd and is a contradiction.

Now suppose that p is odd. Let $k = v_p(n)$. Since this sum is congruent to

$$\frac{n}{p^k} \sum_{x=1}^{p^k} x^{p-1}$$

modulo p^k, it suffices to prove that $p^k \nmid \sum_{x=1}^{p^k} x^{p-1}$. Let g be a primitive root modulo p^k, there exists one by Exercise 3.5.18†. Then,

$$\sum_{x \in (\mathbb{Z}/p\mathbb{Z})^\times} x^{p-1} \equiv \sum_{k=1}^{p^k - 1} g^{k(p-1)} = \frac{g^{p^{k-1}(p-1)^2} - 1}{g^{p-1} - 1}.$$

By LTE 3.4.3, the p-adic valuation of this is $k - 1 < k$. To take care of the terms of the sum which are divisible by p, simply note that

$$v_p \left(\sum_{v_p(x) = \ell, x \in \mathbb{Z}/p^k\mathbb{Z}} x^{p-1} \right) = (p-1)\ell + v_p \left(\sum_{x \in (\mathbb{Z}/p^{k-\ell})^\times} x^{p-1} \right) = (p-1)\ell + k - \ell - 1 \geq k$$

by our previous computation. ∎

Exercise C.5.21 (Finite Fields Kakeya Conjecture, Zeev Dvir)**.** Let $n \geq 1$ an integer and $\mathbb{F}$ a finite field. We say a set $S \subseteq \mathbb{F}^n$ is a *Kakeya set* if it contains a line in every direction, i.e., for every $y \in \mathbb{F}^n$, there exists an $x \in \mathbb{F}^n$ such that S contains the line $x + y\mathbb{F}$. Prove that any polynomial of degree less than $|\mathbb{F}|$ vanishing on a Kakeya set must be zero. Deduce that there is a constant $c_n > 0$ such that, for any finite field $\mathbb{F}$, any Kakeya set of $\mathbb{F}^n$ has cardinality at least $c_n p^n$.

Solution

Let q be the cardinality of $\mathbb{F}$. The proof will be in two steps. Suppose that f is a polynomial of degree $d < q$ vanishing on a Kakeya set S. Fix any $y \in \mathbb{F}^n$. Then, for some $x \in \mathbb{F}^n$, $f(x+ty) = 0$ for any $t \in \mathbb{F}$. The polynomial $f(x+Ty)$ has more roots than its degree so is zero. Let g be the homogeneous part of f, i.e. the polynomial formed by the monomials of degree d of f. Notice that the coefficient of T^d in $f(x+Td)$ is exactly $g(y)$. Hence, $g(y) = 0$ for any $y \in \mathbb{F}^n$, which implies that $g = 0$ by Exercise A.1.7*. This contradicts the assumption that f had degree d.

For the second step, note that the dimension of the vector space V of polynomials of degree at most $q-1$ is $\binom{n+q-1}{n}$. Indeed, the monomials $X_1^{d_1} \cdot \ldots \cdot X_n^{d_n}$ for $d_1 + \ldots + d_n \leq q-1$ form a basis of this space. However, the number of such tuples is the same as the number of ways to choose n elements from $[n+q-1]$: choose $a_1 < \ldots < a_n$ and decide that $d_1 = a_1$, $d_1 + d_2 = a_2$, etc., until $d_1 + \ldots + d_n = a_n$ (this technique is usually called *stars and bars* because you have $q-1$ stars and n bars used to separate them). Now consider the linear map $T : V \to \mathbb{F}^{|S|}$ defined by $T(f) = (f(s))_{s \in S}$. If $|S| < \dim V$, it must have a non-trivial kernel by Proposition C.1.2 (or the rank-nullity theorem). This contradicts the first step.

We conclude that

$$|S| \geq \binom{n+q-1}{n} = \frac{q(q+1) \cdot \ldots \cdot (n+q-1)}{n!} \geq \frac{q^n}{n!}$$

so we can take $c_n = \frac{1}{n!}$. ■

Exercise C.5.22 (Siegel's Lemma)**.** Let $a = (a_{i,j})$ be an $m \times n$ matrix with integer coordinates. Prove that, if $n > m$, the system

$$\sum_{j=1}^{n} a_{i,j} x_j = 0$$

for $i = 1, \ldots, n$ always has a solution in integers with

$$\max_i |x_i| \leq \left(n \max_{i,j} |a_{i,j}| \right)^{\frac{m}{n-m}}.$$

Solution

Let $M = \max_{i,j} |a_{i,j}|$. Fix a constant N to be chosen later. Suppose that the integers $a_1, \ldots, a_k$ are negative while $a_{k+1}, \ldots, a_n$ are positive. Then, for any $(x_1, \ldots, x_n) \in [N]^n$, we have

$$N(a_1 + \ldots + a_k) \leq a_1 x_1 + \ldots + a_n x_n \leq N(a_{k+1} + \ldots + a_n).$$

Thus, the expression $a_1 x_1 + \ldots + a_n x_n$ can take at most $1 + N(|a_1| + \ldots + |a_n|)$ values.

Now, we return to the problem. Set $A = (a_{i,j})$. We have shown that, when $X \subseteq [N]^n$, each rows of AX can take at most $1 + NnM$ values. Thus, when X ranges through $[N]^n$, AX takes at most $(1 + NnM)^m < (1+N)^m (nM)^m$ values. Since X can take $(1+N)^n$ values, if

$$(1+N)^n > (1+N)^m (nM)^m \iff (1+N)^{n-m} > (nM)^m \iff 1 + N > (nM)^{\frac{m}{n-m}}$$

then one value will be taken twice by the pigeonhole principle, say $AX = AY$. This yields

$A(X-Y)=0$ for some $Z=X-Y\subseteq[\![-N,N]\!]^n$ as wanted. It is clear that $N=\left\lfloor (nM)^{\frac{m}{n-m}}\right\rfloor$ works and gives us what we want. ■

Exercise C.5.24. How many invertible $n\times n$ matrices are there in $\mathbb{F}_p$? Deduce the number of (additive) subgroups of cardinality p^m that $(\mathbb{Z}/p\mathbb{Z})^n$ has.

Solution

We proceed inductively to determine the number of tuples of linearly independent vectors of cardinality k. At first, we can pick any non-zero vector in $\mathbb{F}_p^n$, there are thus p^n-1 choices. Then, we can pick any vector which is not a linear combination of the first one, there are thus p^n-p possible choices. Continuing like that, if we have picked k vectors, their linear combinations generate p^k elements so we have p^k elements to avoid and thus p^n-p^k possibilities for the next vector. In conclusion, the number of invertible $n\times n$ matrices with coefficients in $\mathbb{F}_p$ is

$$(p^n-1)(p^n-p)\cdot\ldots\cdot(p^n-p^{n-1}).$$

For the second part, note that a subgroup of $(\mathbb{Z}/p\mathbb{Z})^n$ is a $\mathbb{F}_p$-vector space, and the fact that it has p^m elements means that its dimension is m. Thus, we want to count the subspaces of $\mathbb{F}_p^n$ of dimension m. Here is how we will proceed: we count the number of tuples of m linearly independent elements, and divide this by the number of tuples which represent a fixed subspace. We have already computed the first one: it is

$$(p^n-1)\cdot\ldots\cdot(p^n-p^{m-1}).$$

We have also determined the second: if we fix a subspace of dimension m, it has

$$(p^m-1)\cdot\ldots\cdot(p^m-p^{m-1})$$

bases. We conclude that $(\mathbb{Z}/p\mathbb{Z})^n$ has

$$\frac{(p^n-1)\cdot\ldots\cdot(p^n-p^{m-1})}{(p^m-1)\cdot\ldots\cdot(p^m-p^{m-1})}$$

subgroups of cardinality p^m. ■

Exercise C.5.25†. Let K be a field, and let $S\subseteq K^2$ be a set of points. Prove that there exists a polynomial $f\in K[X,Y]$ of degree at most $\sqrt{2n}$ such that $f(x,y)=0$ for every $(x,y)\in S$.

Solution

Write $f=\sum_{i,j}a_{i,j}X^iY^j$ where the sum is over the i,j such that $i+j\leq\sqrt{2n}$. By stars and bars, there are $\binom{2+\lfloor\sqrt{2n}\rfloor}{2}$ such pairs: we choose the two values i and $i+j+1$ in $[\![0,\lfloor\sqrt{2n}\rfloor+1]\!]$. Since

$$\binom{2+\lfloor\sqrt{2n}\rfloor}{2}=\frac{(1+\lfloor\sqrt{2n}\rfloor)(2+\lfloor\sqrt{2n}\rfloor)}{2}>\frac{\sqrt{2n}\cdot\sqrt{2n}}{2}=n,$$

we have more unknowns then equations so there is a solution. ■

Exercise C.5.26†. Given an $m\times n$ matrix M, we define its *row rank* as the maximal number of linearly independent rows of M. Similarly, its *column rank* is the maximal number of linearly independent columns of M. Prove that these two numbers are the same, called the rank of M and denoted $\operatorname{rank}M$.

Solution

Without loss of generality, by removing some rows if necessary, suppose that all m rows of M are linearly independent. We will prove that M has at least m linearly independent columns, which implies that the column rank is greater than or equal to the row rank (if we add rows the linearly independent columns stay linearly independent). Taking the transpose then yields the reverse inequality, so they are in fact both equal.

Suppose for the sake of a contradiction that M has at most $m-1$ linearly independent columns, say $M^1, \dots, M^k$. If we consider the m vectors corresponding to the rows of $[M^1, \dots, M^k]$, they are linearly dependent by Proposition C.1.2. Now, if we add a column M^{k+1} which is a linear combination of $M^1, \dots, M^k$, the vectors corresponding to the rows of $[M^1, \dots, M^{k+1}]$ stay linearly dependent. Indeed, if

$$M^{k+1} = \sum_{i=1}^{k} a_i M^i$$

and the linear dependence of the rows of $[M^1, \dots, M^k]$ is

$$\sum_{i=1}^{m} b_i m_{i,j}$$

for any $j \in [k]$, then

$$\begin{aligned} \sum_{i=1}^{m} b_i m_{i,k+1} &= \sum_{i=1}^{m} b_i \sum_{j=1}^{k} a_j m_{i,j} \\ &= \sum_{j=1}^{k} a_j \sum_{i=1}^{m} b_i m_{i,j} \\ &= 0. \end{aligned}$$

In other words, a linear dependence between the rows extends to a linear dependence with the same coefficients between the rows when we add a linearly dependent column. Continuing like that shows that, finally, the rows of $[M^1, \dots, M^m] = M$ are linearly dependent, contradicting our initial assumption. ■

Exercise C.5.28 (Nakayama's Lemma)**.** Let R be a commutative ring, I an *ideal* of a R, i.e. an *R-module* inside R[3], and M a finitely-generated R-module. Suppose that $IM = M$, where IM does not mean the set of products of elements of I and M, but instead the R-module it generates (i.e. the set of linear combinations of products). Prove that there exists an element $r \equiv 1 \pmod{I}$ of R such that $rM = 0$.

Solution

Let $\alpha_1, \dots, \alpha_n$ be generators of M. We have a system of equation as follows:

$$\alpha_i = \sum_{j=1}^{n} \beta_{i,j} \alpha_j$$

for $i = 1, \dots, n$ and $\beta_{i,j} \in I$. Let $B = (\beta_{i,j})$ and $A = (\alpha_i)$ (as a colum vector), so that $BA = A$, i.e. $(I_n - B)A = 0$. We claim that $r = \det(I_n - B)$ works. First, note that $r \equiv 1 \pmod{I}$ since $I_n - B \equiv I_n \pmod{I}$. By Proposition C.3.7, we have

$$rI_n = \operatorname{adj}(I_n - B)(I_n - B)$$

[3]See also Proposition C.3.5. A module is like a vector space but the underlying structure is not necessarily a field (in this case it's R).

so, after right-multiplying by A, we get $rA = 0$. This implies that $rM = 0$ as wanted. ■